# Module 4: Cancer

# Team Members: Will Marschall and Dani Folks

# Project Title: Do proliferative-signaling and apoptosis-evasion expression programs track tumor stage in colon adenocarcinoma?

# Project Goal:

Our project seeks to determine whether two cancer hallmarks (**sustained proliferative signaling** and **resisting apoptosis**) show coordinated RNA-expression patterns in **colon adenocarcinoma (COAD)** and whether those patterns may relate to tumor stage.

Our current working hypothesis is that **COAD tumors with higher expression of proliferative-signaling genes will also show stronger apoptosis-resistance patterns and will tend to fall into more advanced tumor-stage groups**, although this remains a hypothesis to be tested over Module 4 rather than a conclusion.

# Disease Background:

## Cancer hallmark focus

For this project, we decided to focus in on two hallmarks that are mechanistically connected and also easy to operationalize with our RNA-expression data:

1. **Sustained growth/proliferative signaling**
   In our quick reference handout, we saw that this hallmark is described as cancer cells no longer waiting for normal external growth signals before dividing. This hallmark increases the cancer-cell division rate and relates idea to overactive growth-factor receptors and downstream signaling genes such as **EGFR, RAS/KRAS, BRAF, and MYC** [1]. In healthy tissue, proliferation is tightly regulated by extracellular ligands (e.g., growth factors) binding to receptors, which activate intracellular signaling cascades only when appropriate. In cancer, this control is bypassed through multiple mechanisms that lead to constitutive activation of growth pathways, including:

   - Overexpression or mutation of receptor tyrosine kinases (e.g., EGFR, ERBB2), allowing signaling even in the absence of ligand
   - Activating mutations in downstream signaling proteins (e.g., KRAS, NRAS, BRAF), which lock pathways such as the MAPK/ERK cascade in an “on” state

- Activation of survival and growth pathways such as PI3K/AKT signaling, promoting both proliferation and metabolic support
- Upregulation of transcription factors (e.g., MYC, FOS, JUN) that directly increase expression of genes required for cell-cycle progression

2. **Resist apoptosis**
   This hallmark describes the ability of cancer cells to evade/resist programmed cell death, a tightly regulated process that normally removes damaged or abnormal cells. In healthy tissue, apoptosis is particularly triggered by DNA damage, oncogenic stress, or loss of survival signals, and is executed through coordinated activation of caspases via intrinsic (mitochondrial) and extrinsic (death-receptor) pathways. For this project, our apoptosis-resistance genes includes **TP53, BCL2, BCL2L1, MCL1, CASP8, CASP3, BAX, BAK1, APAF1, BID, CASP9, FAS, FASLG, BAD, BBC3, BCL2L11, CFLAR, DIABLO, TNF, and TNFRSF10B.** In cancer, this system is disrupted through multiple mechanisms which include:

   - Upregulation of anti-apoptotic proteins **(e.g., BCL2, BCL2L1, MCL1)**, which stabilize mitochondrial membranes and prevent cytochrome c release
   - Downregulation or mutation of pro-apoptotic factors **(e.g., BAX, BAK1, BID)**
   - Loss of tumor suppressor function, especially **TP53**, which we assume would normally promote apoptosis in response to DNA damage
   - Impaired caspase activation, limiting execution of the apoptotic program

   In a functional sense, resisting apoptosis allows cancer cells to survive despite accumulating mutations as well as sustain growth under stress (and during therapy).

## Why we chose colon adenocarcinoma (COAD)

We narrowed our analysis to **one cancer type** instead of many because we were contemplated on deciding whether to focus on one cancer type or to compare multiple cancers, and a single-type analysis is less biologically heterogeneous for our machine-learning workflow discussed in depth during class. We chose **colon adenocarcinoma (COAD)** because:

- Colorectal cancer was emphasized in our 1st lecture and hallmark simulation discussion
- Colon cancer has well-established links to deregulated growth signaling pathways
- Stage information is clinically meaningful for this disease in our initial examination
- One cancer type is a cleaner starting point than attempting many tumor biologies on an initial workflow.
- Colorectal cancer is one of the most common cancers worldwide, ranking third in incidence globally and contributing substantially to cancer-related mortality. [5] This provides us with a strong ethical and scientific motivation to further investigate COAD due to the large number of individuals it affects.

## Overview of hallmark biology

In normal tissues, cells only divide when instructed by extracellular signals, undergo apoptosis when damaged or no longer needed, and remain integrated within the structural and functional organization of the tissue. These processes are largely controlled through coordinated signaling networks that maintain tissue homeostasis and also balance cell proliferation with differentiation and death. Cancer can be easily understood as the breakdown of these regulatory systems, where cells progressively acquire the ability to override multiple biological constraints at the same time. Rather than a single mutation causing disease, tumor development illustrates how the accumulation of changes disrupting normal control mechanisms can allow cells to behave in an erratic manner.

For this project, we focus on two complementary hallmarks: one that acts as the “gas pedal” by driving sustained cell division, and one that weakens the “brake pedal” by impairing programmed cell death, the same analogy used in class to distinguish between oncogenes and tumor suppressors. Together, these hallmarks help us define a two-axis framework in which tumors can be positioned based on how strongly they promote growth versus how effectively they evade death. This paired analysis provides a biologically interpretable way to assess tumor behavior using our provided RNA-expression data.

## Genes associated with the hallmark to be studied

We expanded our hallmark panel to 40 genes total so that the model has a richer feature space while remaining biologically interpretable.

**Sustained proliferative signaling genes** `EGFR, ERBB2, KRAS, NRAS, BRAF, PIK3CA, AKT1, MYC, FOS, JUN, MET, IGF1R, FGFR1, FGFR2, EGF, HGF, SOS1, GRB2, MAPK1, MAPK3`

These genes span many levels of growth signaling pathways, including:

- Receptor-level activation (EGFR, ERBB2)
- Intracellular signal transduction (KRAS, NRAS, BRAF, PIK3CA, AKT1)
- Transcriptional regulation of the cell cycle (MYC, FOS, JUN)

**Resist apoptosis genes** `TP53, BCL2, BCL2L1, MCL1, CASP8, CASP3, BAX, BAK1, APAF1, BID, CASP9, FAS, FASLG, BAD, BBC3, BCL2L11, CFLAR, DIABLO, TNF, TNFRSF10B`

These genes represent key regulators of apoptotic pathways, including:

- Tumor suppressor signaling (TP53)
- Mitochondrial apoptosis regulation (BCL2 family: BCL2, BCL2L1, MCL1, BAX, BAK1, BID)
- Execution of apoptosis via caspases (CASP8, CASP3)

- Apoptosome formation and signaling (APAF1)

A CSV version of our list is included in our repo as `hallmark_gene_sets.csv`. In the EDA step, we explicitly check which of these genes are present in the local training matrix before making any downstream claims.

## Prevalence & incidence

Colorectal cancer is a major public-health burden. The American Cancer Society estimates that in the United States in **2026** there will be about **108,860 new colon cancer cases** and **49,990 new rectal cancer cases**, and colorectal cancer is expected to cause about **55,230 deaths**. Excluding skin cancers, it is the **third most common cancer** diagnosed in both men and women in the United States [5].

## Risk factors (genetic, lifestyle) & societal determinants

Major risk factors for colorectal cancer include **older age**, **family history**, **personal history of colorectal neoplasia**, inherited syndromes such as familial adenomatous polyposis **FAP** and **Lynch syndrome**. Lifestyle-related factors also play a role, including **dietary patterns**, **alcohol consumption**, **cigarette smoking**, and **obesity** [4], [5]. All of which can largely influence inflammation and cellular stress within the colon. **Inflammatory bowel disease (IBD)** further increases risk by repeated cycles of tissue damage and regeneration, which can facilitate mutation accumulation.

In addition to biological risk factors, we note that societal determinants can significantly contribute to risk factors, including access to screening, healthcare availability, socioeconomic status, and geographic location (strongly influence both incidence and outcomes). Early detection can significantly reduce mortality by identifying precancerous lesions or early-stage tumors, whereas limited access to care is associated with later-stage diagnoses and poorer prognosis [5].

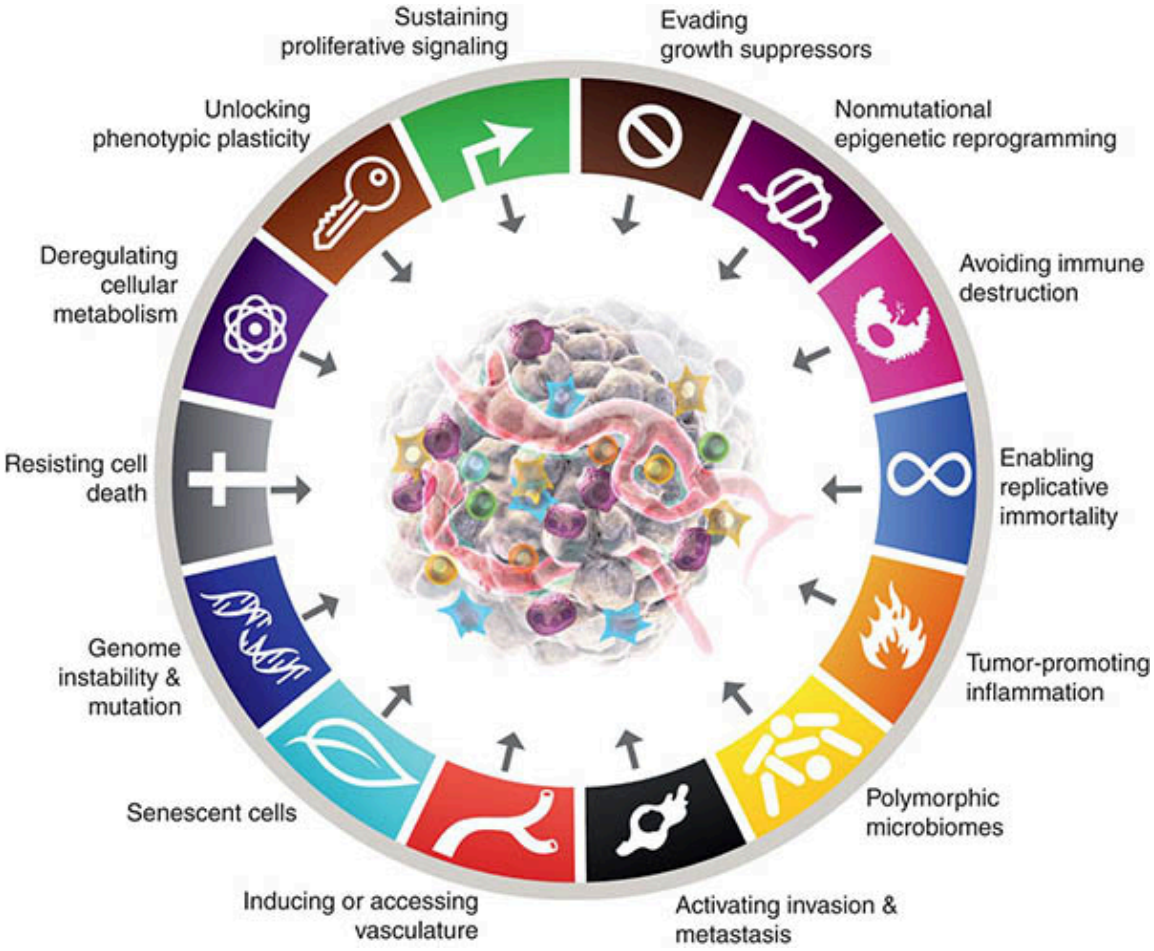
## Standard of care treatments (& reimbursement)

Treatment depends strongly on stage and biomarker status. For early-stage disease, surgical resection is often the primary treatment and can be curative. Standard approaches include **surgery**, **chemotherapy**, **radiation therapy** (more common in rectal disease), **targeted therapy**, and **immunotherapy**.

While we do not make detailed reimbursement claims in this analysis, it is important to note that both tumor stage and biomarker status directly influence treatment selection as well as eligibility for targeted therapies and overall cost of care. This further supports our emphasis on the relevance of linking molecular patterns (such as hallmark-associated gene expression) to clinically meaningful variables like tumor stage.

## Biological mechanisms (anatomy, organ physiology, cell & molecular physiology)

The colon absorbs water and electrolytes from intestinal contents, while the rectum stores stool before elimination. Colon adenocarcinoma arises from epithelial cells in this system. At the molecular level, colorectal tumor development is driven by changes in genes controlling cell division, survival, differentiation, and tissue organization. That makes COAD a good test case for our RNA-based hallmark analysis because aberrant growth signaling and resistance to cell death should both leave measurable transcriptomic signatures for us to interpret [4].




# Data-Set:

The dataset for this project comes from **The Cancer Genome Atlas (TCGA) RNA-seq data reprocessed by Rahman et al.** and distributed through **GEO accession GSE62944** [2], [3]. Dr. Groves further subsetted those data to a smaller teaching dataset containing approximately **50–100 tumors per cancer type** across **24 cancer types** (about **1802 tumors total**) with a reduced metadata table. It should also be noted that not all metadata fields are populated equally across cancer types, so checking missingness is an explicit part of our EDA.

## What these files contain

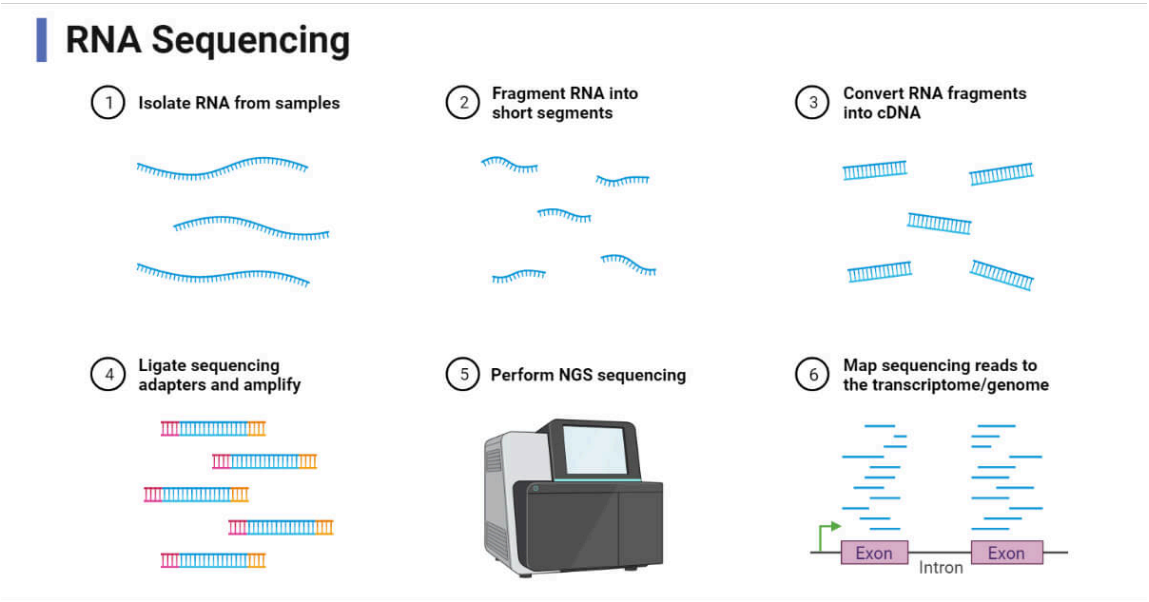
- `TRAINING_SET_GSE62944_subsample_log2TPM.csv`
  Gene-expression matrix with **genes as rows** and **tumor samples as columns**.
- `TRAINING_SET_GSE62944_metadata.csv`
  Metadata table with **samples as rows** and clinical/annotation variables as columns.
- `VALIDATION_SET_...` and `TEST_SET_...`
  We used the **validation set** to compare candidate models and choose our final model, and we used the **test set once at the end** to estimate final out-of-sample performance on completely unseen COAD samples.
- `GSE62944_metadata_percent_nonNA_by_cancer_type.csv`
  A provided summary of metadata completeness by cancer type.

## What is RNA-seq and how should the values be interpreted?

RNA sequencing measures the abundance of RNA transcripts in a sample, which gives a snapshot of our **transcriptome**. In this dataset, the values are **log2(TPM + 1)**:

- **TPM** stands for **transcripts per million**, a normalization that accounts for gene length and sequencing depth.
- The **log2(TPM + 1)** transform compresses the dynamic range so highly expressed genes do not dominate as strongly and zero counts remain defined [2].

Because our dataset is already normalized and log-transformed, it is appropriate for exploratory plots, correlation analysis, and many machine-learning workflows. However, the expression values are **not raw counts**, so count-based methods like DESeq2 would require different input data.




## How the data were generated

Rahman et al. describe reprocessing TCGA RNA-seq data using the **Rsubread** pipeline, aligning reads to the **hg19** reference genome, generating **gene-level integer counts** with `featureCounts`, and then converting those to normalized **FPKM** and **TPM** values. The GEO record states that the dataset contains **9264 tumor samples and 741 normal samples across 24 cancer types** in the full public compendium, along with matched clinical variables [3].

## Metadata we expect to use

The most important of our metadata fields are:

- `cancer_type` to isolate **COAD**
- a **tumor stage** field, if available/populated for COAD in the training split
- possibly age/sex or other basic covariates if available and relevant later on in our analysis

Because stage-column names can differ across datasets, our EDA script searches several likely stage-column names automatically instead of assuming one fixed label.

## What we do in the EDA

Our EDA script is designed to answer these questions:

- What is the size of the training matrix?
- What are the minimum, maximum, and average expression values?
- Which genes have the highest average expression?
- What metadata are missing?
- Are our hallmark genes actually in the dataset?
- How many COAD samples are in the training split?
- Can we visualize the two hallmark programs with scatterplots and PCA?

# Data Analysis:

## Methods

We utilized **unsupervised learning** methods to explore structure in the COAD samples, following the dimensionality-reduction and clustering workflow introduced to us in detail lecture. We first cleaned the metadata by converting age to numeric and simplifying AJCC tumor-stage labels into broader **Stage I**, **Stage II**, **Stage III**, and **Stage IV** bins. We also created a binary stage label (E**arly = Stage I/II**, **Late = Stage III/IV)** for our supervised-learning model.

We then subset the expression matrix to COAD and to a **40-gene hallmark panel**: **20 genes** associated with **sustained proliferative signaling** and **20 genes** associated with **resisting**

**cell death**. Our resulting sample-by-gene matrix was standardized before machine learning so that genes with different expression ranges would contribute more comparably to our downstream analyses.

We used **PCA** as our first dimensionality-reduction method because it provides a linear summary of the strongest axes of variation in the selected hallmark genes. We then used **K-means clustering** to look for exploratory subgroups in the standardized gene-expression matrix. K-means minimizes the within-cluster distance between samples and their assigned centroids, so it provides us with a great and simple first pass at identifying molecular groupings. We evaluated candidate values of k from **2 to 6** using silhouette score and selected the **best-performing k**. We also used **UMAP** as a nonlinear dimensionality-reduction method to compare whether local neighborhood structure looked similar to our PCA view.

For supervised learning, we used **logistic regression** to classify simplified tumor stage (**Early = Stage I/II**, **Late = Stage III/IV**) from hallmark-expression features in COAD. We chose logistic regression because our target is binary, our sample size is modest, and the model provides interpretable coefficients while also supporting regularization.

Following the workflow discussed in class, we used the **training dataset** to fit candidate models, the **validation dataset** to compare models and judge generalization, and the **test dataset only once at the end** to estimate final out-of-sample performance on unseen data. Our baseline model used the 40 hallmark genes alone. Our improved model added biologically meaningful summary features (**proliferative score**, **apoptosis-resistance score**, and **age at diagnosis**) and used class balancing plus stronger regularization to reduce overfitting.

To evaluate performance, we used both **threshold-dependent** and **threshold-independent** metrics. At the default threshold of 0.5, we reported **accuracy**, **balanced accuracy**, **precision**, **recall**, **F1 score**, and the **confusion matrix**. Across all thresholds, we reported **AUROC**. After selecting the better-performing model using validation performance, we applied that final model to the **held-out test set** and reported the final out-of-sample metrics.

# Analysis

## 1. Set up the notebook and import our analysis script

In this first step, we set up the notebook environment and import our main.py script from the code folder. This lets us keep the full analysis logic in one reusable Python file while still showing the major workflow steps inside the notebook.

In [1]:
```
# This is a short snippet from our main.py script.
# The full script is quite long, so we are only showing the top portion here where
# Please view our main.py script for full detailed comments, methodologies, and res
```

```
from pathlib import Path
import sys
import importlib
import pandas as pd
from IPython.display import display, Image

pd.options.display.float_format = "{:.6f}".format

NOTEBOOK_DIR = Path.cwd()
PROJECT_ROOT = NOTEBOOK_DIR.parent if NOTEBOOK_DIR.name.lower() == "code" else NOTE
CODE_DIR = PROJECT_ROOT / "code"

if str(CODE_DIR) not in sys.path:
    sys.path.append(str(CODE_DIR))

import main
importlib.reload(main)
```

Out[1]:

```
<module 'main' from 'e:\\OneDrive - University of Virginia\\Will M\\Undergraduate
Research Courses-Grants\\05_Spring_2026\\Computational_Biomedical_BME_2315\\Module
-4-Cancer\\code\\main.py'>
```

## 2. Run the full exploratory workflow on the COAD training set

This cell runs the full exploratory workflow defined in `main.py`. It loads the training expression matrix and metadata, subsets the data to COAD, checks whether the selected hallmark genes are present, cleans the metadata, computes hallmark scores, performs PCA and K-means clustering, and generates the summary figures saved to the `results` folder.

In [2]:

```
# We wrap the entire EDA workflow in a function and call it in the main block.
# This shows our entire EDA workflow in one place and allows us to easily reuse the
eda_outputs = main.run_eda()
```

```
Loaded expression data.
        TCGA-E9-A1NI-01A-11R-A14D-07  TCGA-E2-A1LK-01A-21R-A14D-07  \
A1BG                        3.397369                      3.466089
A1CF                        0.008857                      0.039562
A2M                         7.575125                      6.643613
A2ML1                       0.397610                      7.625124
A4GALT                      5.277425                      5.244677

        TCGA-BH-A0B2-01A-11R-A10J-07  TCGA-E2-A107-01A-11R-A10J-07  \
A1BG                        3.789771                      3.967578
A1CF                        0.065051                      0.000000
A2M                         9.024479                      7.573842
A2ML1                       0.428689                      0.465410
A4GALT                      4.072650                      4.208381

        TCGA-LL-A5YN-01A-11R-A28M-07  TCGA-BH-A0DQ-01A-11R-A084-07  \
A1BG                        4.733007                      3.011343
A1CF                        0.014260                      0.008014
A2M                         7.459105                      9.279760
A2ML1                       1.126008                      0.083225
A4GALT                      5.249234                      5.123996

        TCGA-D8-A73X-01A-11R-A32P-07  TCGA-AR-A0TP-01A-11R-A084-07  \
A1BG                        4.324578                      3.687565
A1CF                        0.021112                      0.020984
A2M                         7.526283                     12.770294
A2ML1                       0.274456                      0.220169
A4GALT                      5.465881                      3.083101

        TCGA-E2-A1IF-01A-11R-A144-07  TCGA-EW-A6SD-01A-12R-A33J-07  ...  \
A1BG                        2.399717                      4.233443  ...
A1CF                        0.025614                      0.108061  ...
A2M                         8.201515                      8.065138  ...
A2ML1                       0.152375                      1.333526  ...
A4GALT                      5.371876                      3.027163  ...

        TCGA-N5-A4RF-01A-11R-A28V-07  TCGA-N6-A4VF-01A-31R-A28V-07  \
A1BG                        4.434854                      3.630666
A1CF                        0.044659                      0.007317
A2M                         9.212672                     10.137570
A2ML1                       0.063074                      0.546730
A4GALT                      4.821015                      5.287393

        TCGA-N5-A4RN-01A-12R-A28V-07  TCGA-QM-A5NM-01A-11R-A28V-07  \
A1BG                        3.575404                      2.458971
A1CF                        0.015572                      0.035358
A2M                         8.439769                      5.508460
A2ML1                       0.744056                      2.409112
A4GALT                      4.690114                      4.438399

        TCGA-N5-A4RJ-01A-11R-A28V-07  TCGA-N5-A4RO-01A-11R-A28V-07  \
A1BG                        5.791871                      3.535440
A1CF                        1.949092                      0.078157
A2M                         6.958057                      7.928136
A2ML1                       0.686557                      0.088065
A4GALT                      4.914086                      3.492874
```

```
        TCGA-N5-A4RV-01A-21R-A28V-07  TCGA-N6-A4VD-01A-11R-A28V-07  \
A1BG                        6.075494                      4.383862
A1CF                        0.021624                      0.006336
A2M                         9.668470                      8.952802
A2ML1                       3.002216                      0.977974
A4GALT                      2.089483                      5.706015

        TCGA-N5-A4RT-01A-11R-A28V-07  TCGA-ND-A4WC-01A-21R-A28V-07
A1BG                        4.488649                      3.182423
A1CF                        0.019464                      0.091276
A2M                         6.137774                      7.317486
A2ML1                       0.568217                      0.012821
A4GALT                      4.227761                      5.491313

[5 rows x 1802 columns]

Expression matrix shape (genes x samples): (15716, 1802)

Metadata shape (samples x features): (1802, 29)

Expression matrix info:
<class 'pandas.core.frame.DataFrame'>
Index: 15716 entries, A1BG to ZZZ3
Columns: 1802 entries, TCGA-E9-A1NI-01A-11R-A14D-07 to TCGA-ND-A4WC-01A-21R-A28V-07
dtypes: float64(1802)
memory usage: 216.2+ MB
None

Metadata info:
<class 'pandas.core.frame.DataFrame'>
Index: 1802 entries, TCGA-E9-A1NI-01A-11R-A14D-07 to TCGA-ND-A4WC-01A-21R-A28V-07
Data columns (total 29 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   cancer_type                           1802 non-null   object
 1   bcr_patient_barcode                   1730 non-null   object
 2   gender                                1730 non-null   object
 3   race                                  1730 non-null   object
 4   ethnicity                             1730 non-null   object
 5   age_at_diagnosis                      1730 non-null   object
 6   ajcc_pathologic_tumor_stage           1144 non-null   object
 7   ajcc_nodes_pathologic_pn              1142 non-null   object
 8   ajcc_metastasis_pathologic_pm         1142 non-null   object
 9   ajcc_tumor_pathologic_pt              1142 non-null   object
 10  tumor_status                          1650 non-null   object
 11  histologic_diagnosis                  1272 non-null   object
 12  history_other_malignancy              1730 non-null   object
 13  history_thyroid_disease               79 non-null     object
 14  history_thyroid_disease_other         79 non-null     object
 15  history_reflux_disease_indicator      80 non-null     object
 16  history_hormonal_contraceptives_use   211 non-null    object
 17  pregnancies_count_miscarriage         77 non-null     object
 18  jewish_religion_heritage_indicator    77 non-null     object
 19  tobacco_smoking_age_started           153 non-null    object
 20  family_history_cancer_type            76 non-null     object
```

```
 21  OS                                  1727 non-null   float64
 22  OS.time                             1719 non-null   float64
 23  DSS                                 1625 non-null   float64
 24  DSS.time                            1640 non-null   float64
 25  DFI                                 813 non-null    float64
 26  DFI.time                            812 non-null    float64
 27  PFI                                 1647 non-null   float64
 28  PFI.time                            1642 non-null   float64
dtypes: float64(8), object(21)
memory usage: 422.3+ KB
None

Global expression summary:
   n_genes  n_samples  global_min_log2TPMplus1  global_max_log2TPMplus1  \
0    15716       1802                 0.000000                17.410280

   global_mean_log2TPMplus1  global_median_log2TPMplus1
0                  3.716808                    3.809865

Top 20 genes by average expression across the training set:
        mean_expression
ACTB          12.594034
FTL           12.525603
TPT1          12.476975
B2M           12.347839
GAPDH         12.160252
TMSB10        11.998153
RPS11         11.933026
ACTG1         11.866736
RPS6          11.836285
RPL8          11.655261
RPLP1         11.627998
RPL30         11.564944
RPL37A        11.405454
RPL18         11.291977
RPL11         11.282330
RPLP2         11.274173
GNB2L1        11.264597
RPS8          11.264118
RPS4X         11.195749
RPS16         11.060553

Metadata missingness summary:
                              n_non_missing  n_missing  percent_non_missing  \
cancer_type                            1802          0           100.000000
history_other_malignancy               1730         72            96.004440
gender                                 1730         72            96.004440
race                                   1730         72            96.004440
ethnicity                              1730         72            96.004440
age_at_diagnosis                       1730         72            96.004440
bcr_patient_barcode                    1730         72            96.004440
OS                                     1727         75            95.837958
OS.time                                1719         83            95.394007
tumor_status                           1650        152            91.564928
PFI                                    1647        155            91.398446
PFI.time                               1642        160            91.120977
```

```
DSS.time                                  1640        162            91.009989
DSS                                       1625        177            90.177580
histologic_diagnosis                      1272        530            70.588235
ajcc_pathologic_tumor_stage               1144        658            63.485017
ajcc_tumor_pathologic_pt                  1142        660            63.374029
ajcc_metastasis_pathologic_pm             1142        660            63.374029
ajcc_nodes_pathologic_pn                  1142        660            63.374029
DFI                                        813        989            45.116537

                                 percent_missing
cancer_type                             0.000000
history_other_malignancy                3.995560
gender                                  3.995560
race                                    3.995560
ethnicity                               3.995560
age_at_diagnosis                        3.995560
bcr_patient_barcode                     3.995560
OS                                      4.162042
OS.time                                 4.605993
tumor_status                            8.435072
PFI                                     8.601554
PFI.time                                8.879023
DSS.time                                8.990011
DSS                                     9.822420
histologic_diagnosis                   29.411765
ajcc_pathologic_tumor_stage            36.514983
ajcc_tumor_pathologic_pt               36.625971
ajcc_metastasis_pathologic_pm          36.625971
ajcc_nodes_pathologic_pn               36.625971
DFI                                    54.883463

Subset to COAD:
Expression shape (genes x selected samples): (15716, 80)
Metadata shape (selected samples x features): (80, 29)

Simplified stage counts:
stage_simple
Stage II     30
Stage I      17
Stage III    16
<NA>         10
Stage IV      7
Name: count, dtype: int64

Binary stage counts (planned future supervised target):
stage_binary
Early    47
Late     23
NaN      10
Name: count, dtype: int64

Requested hallmark genes and whether they were found:
                              hallmark       gene  present_in_dataset
0   Sustained_Proliferative_Signaling       EGFR                True
1   Sustained_Proliferative_Signaling      ERBB2                True
2   Sustained_Proliferative_Signaling       KRAS                True
```

| | | | |
|---|---|---|---|
| 3 | Sustained_Proliferative_Signaling | NRAS | True |
| 4 | Sustained_Proliferative_Signaling | BRAF | True |
| 5 | Sustained_Proliferative_Signaling | PIK3CA | True |
| 6 | Sustained_Proliferative_Signaling | AKT1 | True |
| 7 | Sustained_Proliferative_Signaling | MYC | True |
| 8 | Sustained_Proliferative_Signaling | FOS | True |
| 9 | Sustained_Proliferative_Signaling | JUN | True |
| 10 | Sustained_Proliferative_Signaling | MET | True |
| 11 | Sustained_Proliferative_Signaling | IGF1R | True |
| 12 | Sustained_Proliferative_Signaling | FGFR1 | True |
| 13 | Sustained_Proliferative_Signaling | FGFR2 | True |
| 14 | Sustained_Proliferative_Signaling | EGF | True |
| 15 | Sustained_Proliferative_Signaling | HGF | True |
| 16 | Sustained_Proliferative_Signaling | SOS1 | True |
| 17 | Sustained_Proliferative_Signaling | GRB2 | True |
| 18 | Sustained_Proliferative_Signaling | MAPK1 | True |
| 19 | Sustained_Proliferative_Signaling | MAPK3 | True |
| 20 | Resist_Apoptosis | TP53 | True |
| 21 | Resist_Apoptosis | BCL2 | True |
| 22 | Resist_Apoptosis | BCL2L1 | True |
| 23 | Resist_Apoptosis | MCL1 | True |
| 24 | Resist_Apoptosis | CASP8 | True |
| 25 | Resist_Apoptosis | CASP3 | True |
| 26 | Resist_Apoptosis | BAX | True |
| 27 | Resist_Apoptosis | BAK1 | True |
| 28 | Resist_Apoptosis | APAF1 | True |
| 29 | Resist_Apoptosis | BID | True |
| 30 | Resist_Apoptosis | CASP9 | True |
| 31 | Resist_Apoptosis | FAS | True |
| 32 | Resist_Apoptosis | FASLG | True |
| 33 | Resist_Apoptosis | BAD | True |
| 34 | Resist_Apoptosis | BBC3 | True |
| 35 | Resist_Apoptosis | BCL2L11 | True |
| 36 | Resist_Apoptosis | CFLAR | True |
| 37 | Resist_Apoptosis | DIABLO | True |
| 38 | Resist_Apoptosis | TNF | True |
| 39 | Resist_Apoptosis | TNFRSF10B | True |

Hallmark gene expression subset:

| | TCGA-AA-3532-01A-01R-0821-07 | TCGA-AA-3842-01A-01R-1022-07 \ |
|---|---|---|
| EGFR | 2.944719 | 3.345953 |
| ERBB2 | 6.589719 | 5.959237 |
| KRAS | 4.427641 | 5.057217 |
| NRAS | 6.095053 | 6.081736 |
| BRAF | 2.697256 | 2.308653 |

| | TCGA-G4-6310-01A-11R-1723-07 | TCGA-AA-3975-01A-01R-1022-07 \ |
|---|---|---|
| EGFR | 3.880086 | 3.238562 |
| ERBB2 | 7.155463 | 7.570348 |
| KRAS | 4.527352 | 3.957377 |
| NRAS | 5.789315 | 5.514185 |
| BRAF | 1.985613 | 2.859851 |

| | TCGA-AM-5820-01A-01R-1653-07 | TCGA-AA-A02W-01A-01R-A00A-07 \ |
|---|---|---|
| EGFR | 4.524024 | 3.511442 |
| ERBB2 | 6.676083 | 6.895618 |

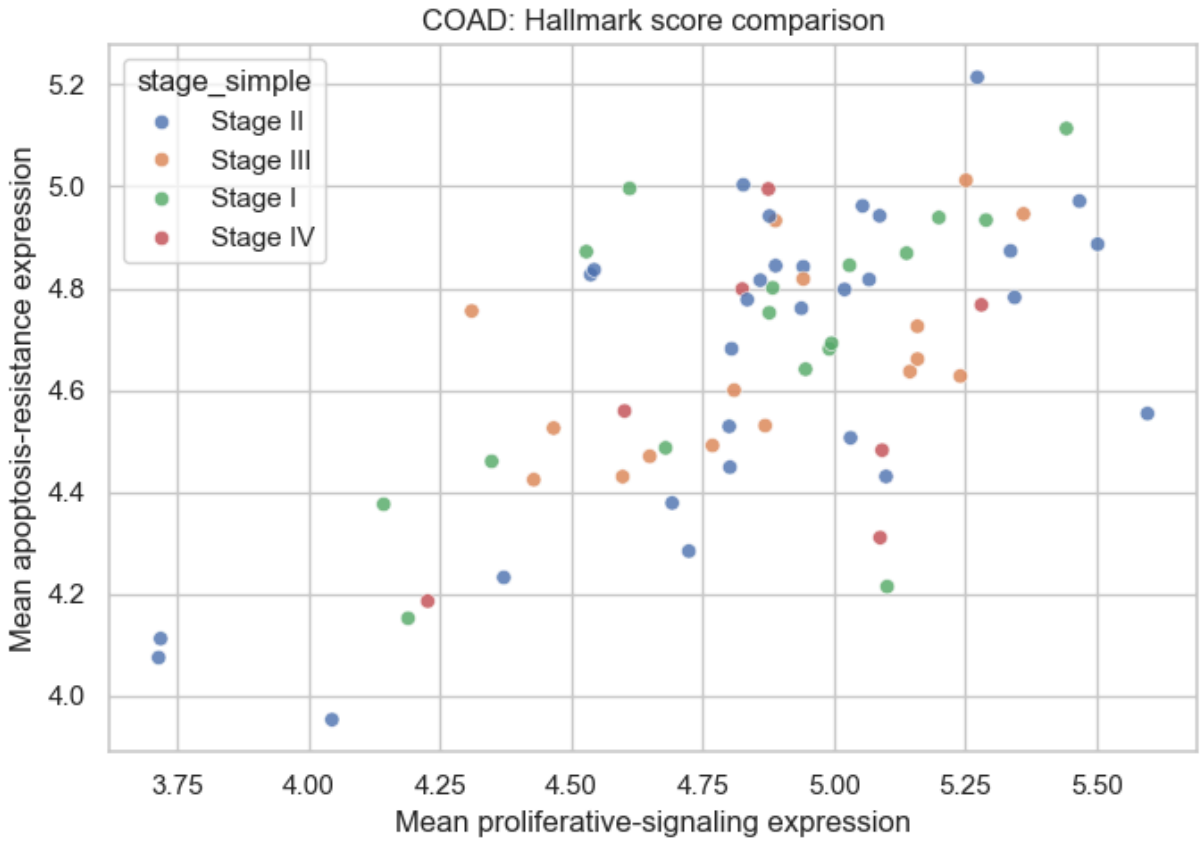
```
KRAS                       5.339606                     4.941172
NRAS                       6.263340                     5.843279
BRAF                       3.011341                     2.336622

       TCGA-CA-6719-01A-11R-1839-07  TCGA-G4-6295-01A-11R-1723-07  \
EGFR                       5.067206                     3.775271
ERBB2                      7.944244                     5.938120
KRAS                       4.691026                     4.755830
NRAS                       5.842546                     6.103702
BRAF                       2.611146                     1.889755

       TCGA-AA-3949-01A-01R-1022-07  TCGA-AA-3815-01A-01R-1022-07  ...  \
EGFR                       2.396876                     2.962133  ...
ERBB2                      5.799920                     7.153184  ...
KRAS                       5.292536                     4.294864  ...
NRAS                       5.791701                     5.247356  ...
BRAF                       1.982464                     1.989730  ...

       TCGA-A6-2674-01B-04R-A277-07  TCGA-D5-6533-01A-11R-1723-07  \
EGFR                       5.310483                     4.039504
ERBB2                      4.166716                     6.167350
KRAS                       7.358750                     4.933434
NRAS                       6.930299                     5.542449
BRAF                       5.056716                     2.486954

       TCGA-AA-3549-01A-02R-0821-07  TCGA-AA-3667-01A-01R-0905-07  \
EGFR                       2.812521                     2.987640
ERBB2                      6.163905                     7.252683
KRAS                       4.365847                     2.468993
NRAS                       5.002958                     3.566642
BRAF                       1.595834                     1.639860

       TCGA-AA-3854-01A-01R-0905-07  TCGA-AA-3562-01A-02R-0821-07  \
EGFR                       3.720839                     3.444506
ERBB2                      7.197813                     6.722895
KRAS                       5.120718                     3.958116
NRAS                       5.403561                     5.843603
BRAF                       2.688802                     1.927858

       TCGA-AA-3655-01A-02R-1723-07  TCGA-F4-6463-01A-11R-1723-07  \
EGFR                       4.119861                     3.302235
ERBB2                      7.520330                     7.002199
KRAS                       5.384852                     5.251559
NRAS                       6.006198                     5.739382
BRAF                       2.200716                     1.805409

       TCGA-AA-A01R-01A-21R-A083-07  TCGA-A6-5657-01A-01R-A32Z-07
EGFR                       3.333926                     4.253522
ERBB2                      7.177614                     6.873403
KRAS                       3.131266                     4.480993
NRAS                       3.618053                     5.666606
BRAF                       1.720988                     2.081668

[5 rows x 80 columns]

Hallmark gene statistics within the selected cancer type:
```

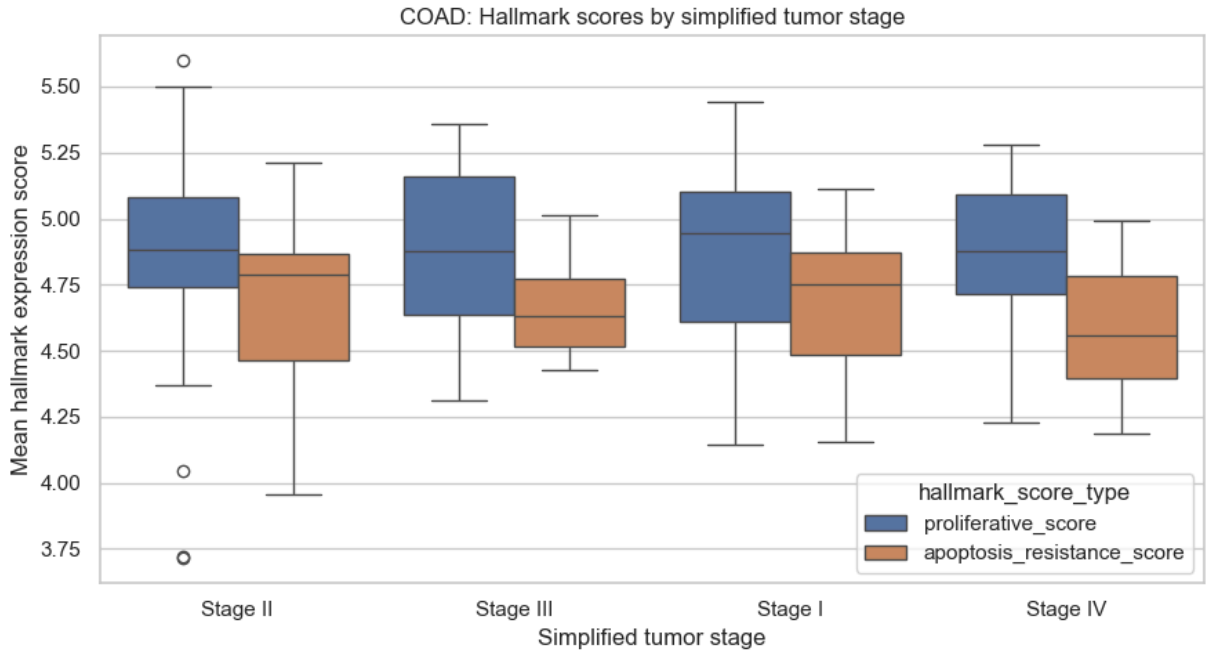
| | mean_expression | median_expression | variance_across_samples |
|---|---|---|---|
| MYC | 8.228511 | 8.221083 | 0.659806 |
| FOS | 7.957519 | 7.856431 | 1.629929 |
| JUN | 7.618991 | 7.732972 | 0.609185 |
| MCL1 | 7.431396 | 7.501133 | 0.490321 |
| BCL2L1 | 7.398317 | 7.387813 | 0.527915 |
| MAPK3 | 7.251998 | 7.294059 | 0.606089 |
| BAX | 7.229717 | 7.268432 | 0.492506 |
| AKT1 | 7.080532 | 7.166449 | 0.234295 |
| GRB2 | 6.810953 | 6.816642 | 0.198127 |
| ERBB2 | 6.782761 | 6.903818 | 0.675691 |
| TP53 | 6.080440 | 6.598347 | 1.795712 |
| BAD | 6.038458 | 6.123912 | 0.308288 |
| DIABLO | 5.965331 | 5.964623 | 0.111331 |
| MET | 5.942898 | 5.879211 | 0.717186 |
| BID | 5.829742 | 5.781805 | 0.251465 |
| MAPK1 | 5.705211 | 5.766001 | 0.310708 |
| BAK1 | 5.660182 | 5.656553 | 0.338468 |
| TNFRSF10B | 5.576977 | 5.615052 | 0.548658 |
| CASP3 | 5.522967 | 5.678818 | 0.371093 |
| NRAS | 5.477788 | 5.670852 | 0.745258 |
| BBC3 | 4.617883 | 4.658929 | 0.410255 |
| KRAS | 4.602230 | 4.672501 | 0.763439 |
| CASP8 | 4.347573 | 4.397027 | 0.343758 |
| CFLAR | 3.926365 | 3.904053 | 0.406354 |
| SOS1 | 3.791937 | 3.822083 | 0.534900 |
| CASP9 | 3.782139 | 3.823212 | 0.258526 |
| IGF1R | 3.769224 | 3.701273 | 0.551881 |
| FAS | 3.736817 | 3.694270 | 1.060056 |
| EGFR | 3.470916 | 3.455831 | 0.627900 |
| APAF1 | 3.358721 | 3.413749 | 0.507264 |
| FGFR1 | 3.284186 | 3.192755 | 1.169687 |
| BCL2L11 | 3.273790 | 3.326452 | 0.400095 |
| PIK3CA | 3.229337 | 3.308622 | 0.627246 |
| FGFR2 | 2.575545 | 2.484944 | 1.273997 |
| BRAF | 2.216751 | 2.186404 | 0.426419 |
| BCL2 | 1.938258 | 1.990227 | 0.795100 |
| FASLG | 0.970790 | 0.775964 | 0.546242 |
| HGF | 0.968300 | 0.855008 | 0.324848 |
| EGF | 0.881202 | 0.338483 | 1.149523 |
| TNF | 0.295415 | 0.247102 | 0.047064 |

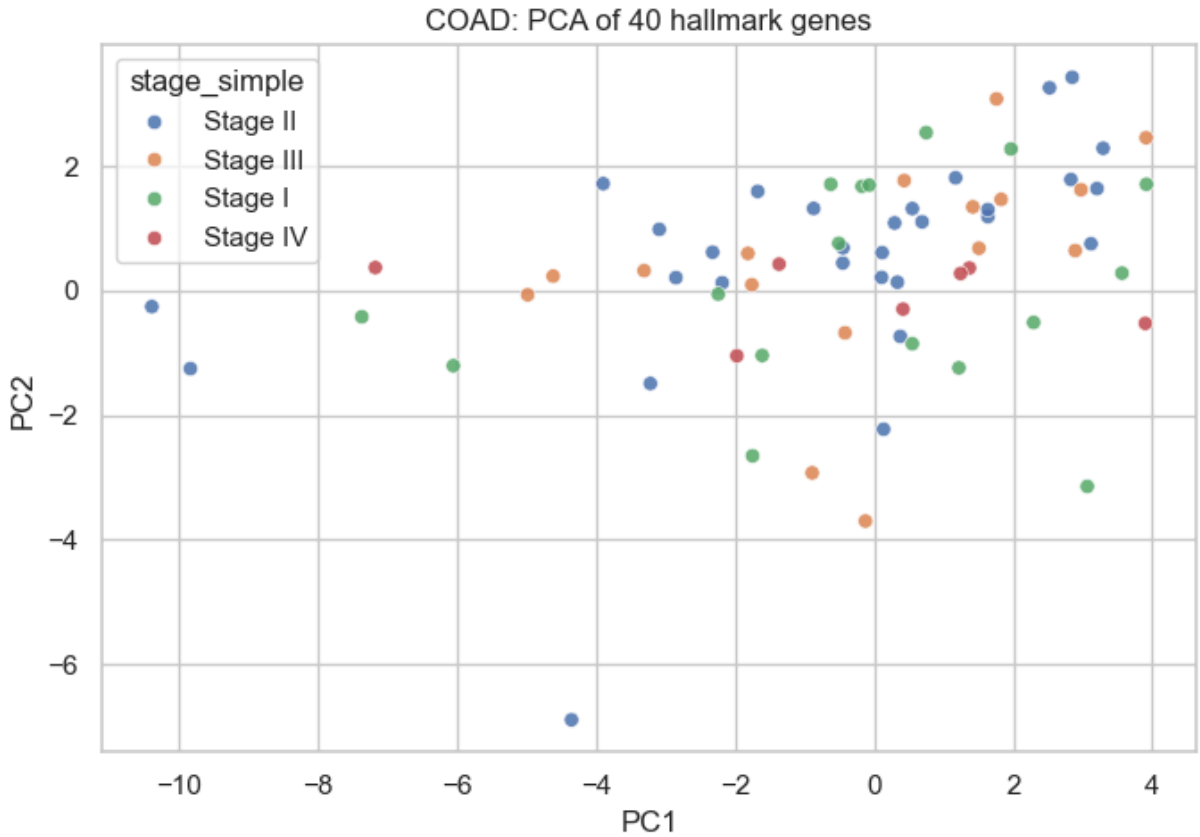
```
Using metadata stage column: ajcc_pathologic_tumor_stage
ajcc_pathologic_tumor_stage
Stage IIA          24
Stage I            17
NaN                 8
Stage IIIB          7
Stage II            5
Stage IV            5
Stage IIIC          5
Stage III           3
Stage IVA           2
[Not Available]     2
Stage IIIA          1
Stage IIB           1
```

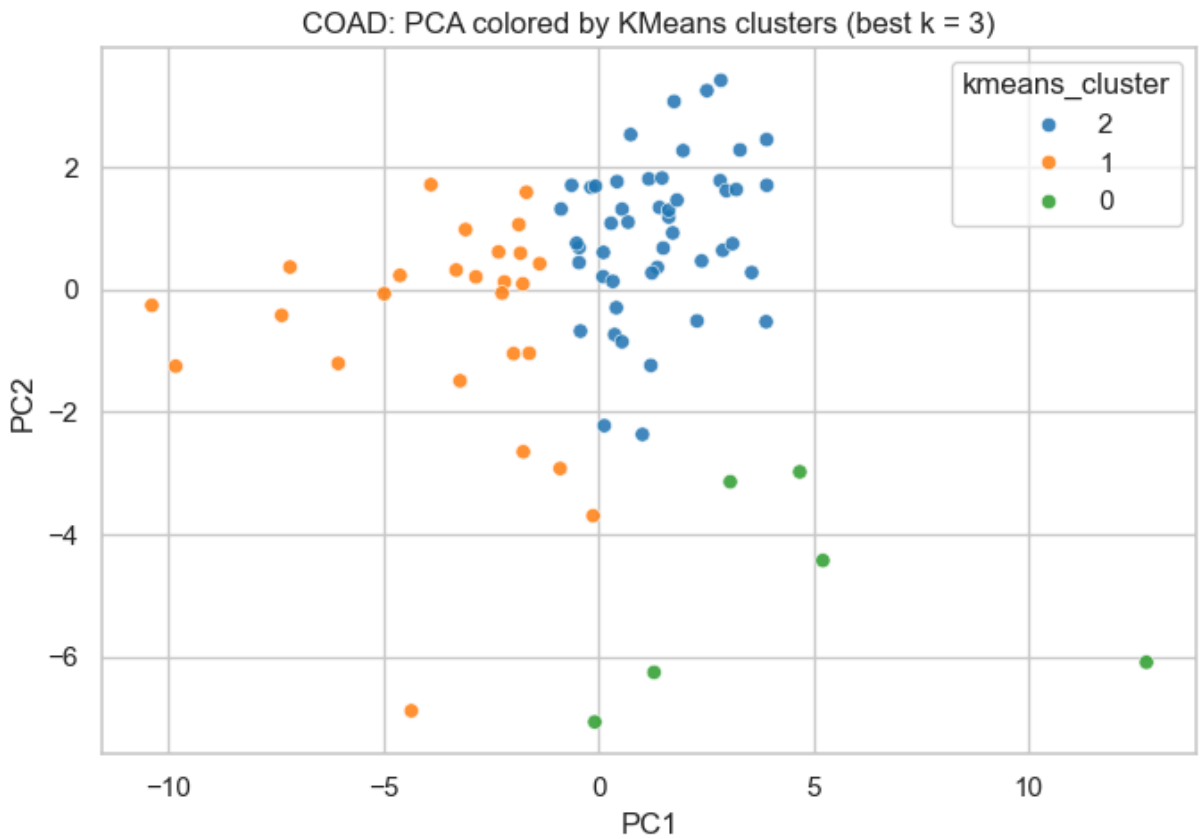
```
Name: count, dtype: int64

Sample-level hallmark score table:
                              proliferative_score  apoptosis_resistance_score  \
TCGA-AA-3532-01A-01R-0821-07             4.805219                    4.681239
TCGA-AA-3842-01A-01R-1022-07             4.869162                    4.530510
TCGA-G4-6310-01A-11R-1723-07             5.144955                    4.636565
TCGA-AA-3975-01A-01R-1022-07             4.991320                    4.681024
TCGA-AM-5820-01A-01R-1653-07             5.280900                    4.767320

                             cancer_type bcr_patient_barcode  gender  \
TCGA-AA-3532-01A-01R-0821-07        COAD        TCGA-AA-3532    MALE
TCGA-AA-3842-01A-01R-1022-07        COAD        TCGA-AA-3842    MALE
TCGA-G4-6310-01A-11R-1723-07        COAD        TCGA-G4-6310    MALE
TCGA-AA-3975-01A-01R-1022-07        COAD        TCGA-AA-3975    MALE
TCGA-AM-5820-01A-01R-1653-07        COAD        TCGA-AM-5820  FEMALE

                                          race               ethnicity  \
TCGA-AA-3532-01A-01R-0821-07  [Not Available]         [Not Available]
TCGA-AA-3842-01A-01R-1022-07  [Not Available]         [Not Available]
TCGA-G4-6310-01A-11R-1723-07            WHITE  NOT HISPANIC OR LATINO
TCGA-AA-3975-01A-01R-1022-07  [Not Available]         [Not Available]
TCGA-AM-5820-01A-01R-1653-07            WHITE  NOT HISPANIC OR LATINO

                              age_at_diagnosis ajcc_pathologic_tumor_stage  \
TCGA-AA-3532-01A-01R-0821-07         63.000000                   Stage IIA
TCGA-AA-3842-01A-01R-1022-07         51.000000                  Stage IIIA
TCGA-G4-6310-01A-11R-1723-07         69.000000                  Stage IIIB
TCGA-AA-3975-01A-01R-1022-07         80.000000                     Stage I
TCGA-AM-5820-01A-01R-1653-07         59.000000                   Stage IVA

                             ajcc_nodes_pathologic_pn  ...     OS.time  \
TCGA-AA-3532-01A-01R-0821-07                       N0  ...  882.000000
TCGA-AA-3842-01A-01R-1022-07                       N1  ... 1126.000000
TCGA-G4-6310-01A-11R-1723-07                       N1  ... 1935.000000
TCGA-AA-3975-01A-01R-1022-07                       N0  ... 1036.000000
TCGA-AM-5820-01A-01R-1653-07                       N2  ...   14.000000

                                 DSS    DSS.time      DFI    DFI.time  \
TCGA-AA-3532-01A-01R-0821-07 0.000000  882.000000      NaN         NaN
TCGA-AA-3842-01A-01R-1022-07 0.000000 1126.000000 1.000000  396.000000
TCGA-G4-6310-01A-11R-1723-07 0.000000 1935.000000 0.000000 1935.000000
TCGA-AA-3975-01A-01R-1022-07 0.000000 1036.000000 0.000000 1036.000000
TCGA-AM-5820-01A-01R-1653-07 0.000000   14.000000      NaN         NaN

                                 PFI    PFI.time stage_clean stage_simple  \
TCGA-AA-3532-01A-01R-0821-07 0.000000  882.000000   Stage IIA     Stage II
TCGA-AA-3842-01A-01R-1022-07 1.000000  396.000000  Stage IIIA    Stage III
TCGA-G4-6310-01A-11R-1723-07 0.000000 1935.000000  Stage IIIB    Stage III
TCGA-AA-3975-01A-01R-1022-07 0.000000 1036.000000     Stage I      Stage I
TCGA-AM-5820-01A-01R-1653-07 0.000000   14.000000   Stage IVA     Stage IV

                             stage_binary
TCGA-AA-3532-01A-01R-0821-07        Early
TCGA-AA-3842-01A-01R-1022-07         Late
TCGA-G4-6310-01A-11R-1723-07         Late
```

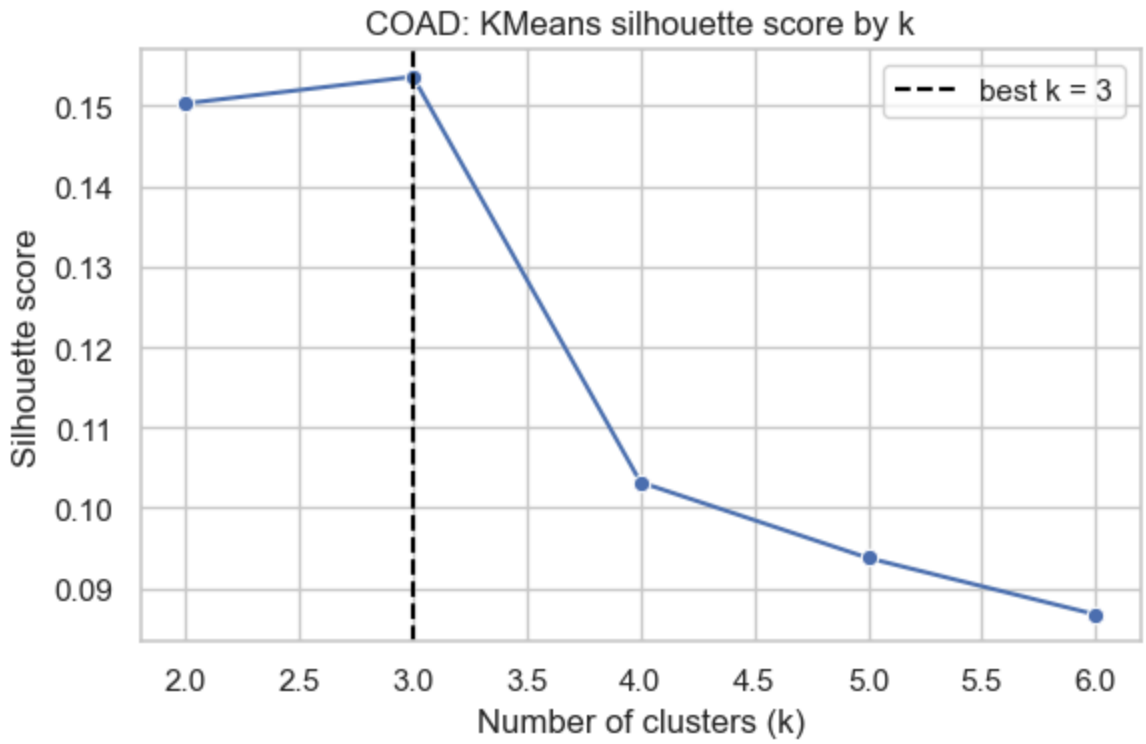
```
TCGA-AA-3975-01A-01R-1022-07        Early
TCGA-AM-5820-01A-01R-1653-07         Late

[5 rows x 34 columns]
e:\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans i
s known to have a memory leak on Windows with MKL, when there are less chunks than a
vailable threads. You can avoid it by setting the environment variable OMP_NUM_THREA
DS=1.
  warnings.warn(
e:\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans i
s known to have a memory leak on Windows with MKL, when there are less chunks than a
vailable threads. You can avoid it by setting the environment variable OMP_NUM_THREA
DS=1.
  warnings.warn(
e:\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans i
s known to have a memory leak on Windows with MKL, when there are less chunks than a
vailable threads. You can avoid it by setting the environment variable OMP_NUM_THREA
DS=1.
  warnings.warn(
e:\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans i
s known to have a memory leak on Windows with MKL, when there are less chunks than a
vailable threads. You can avoid it by setting the environment variable OMP_NUM_THREA
DS=1.
  warnings.warn(
e:\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans i
s known to have a memory leak on Windows with MKL, when there are less chunks than a
vailable threads. You can avoid it by setting the environment variable OMP_NUM_THREA
DS=1.
  warnings.warn(
e:\Anaconda\Lib\site-packages\umap\umap_.py:1952: UserWarning: n_jobs value 1 overri
dden to 1 by setting random_state. Use no seed for parallelism.
  warn(
KMeans model selection summary:
   k     inertia  silhouette_score
0  2 2682.814190          0.150278
1  3 2411.957810          0.153595
2  4 2189.351156          0.103196
3  5 2057.638214          0.093785
4  6 1957.743396          0.086763

Best KMeans k based on silhouette score: 3 (silhouette = 0.154)

Copied metadata completeness table to results folder.
```

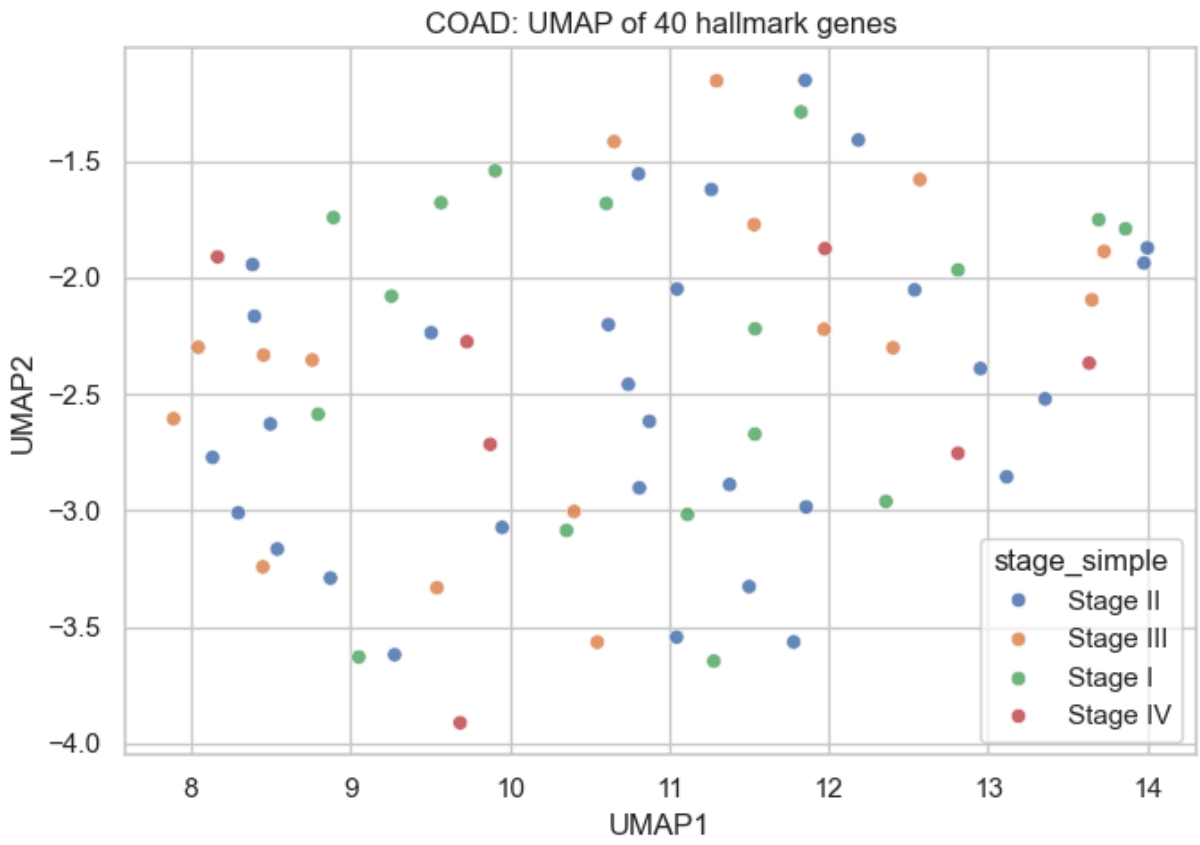

COAD: Hallmark score comparison
stage_simple
Stage II
Stage III
Stage I
Stage IV
Mean apoptosis-resistance expression
Mean proliferative-signaling expression
5.2
5.0
4.8
4.6
4.4
4.2
4.0
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50

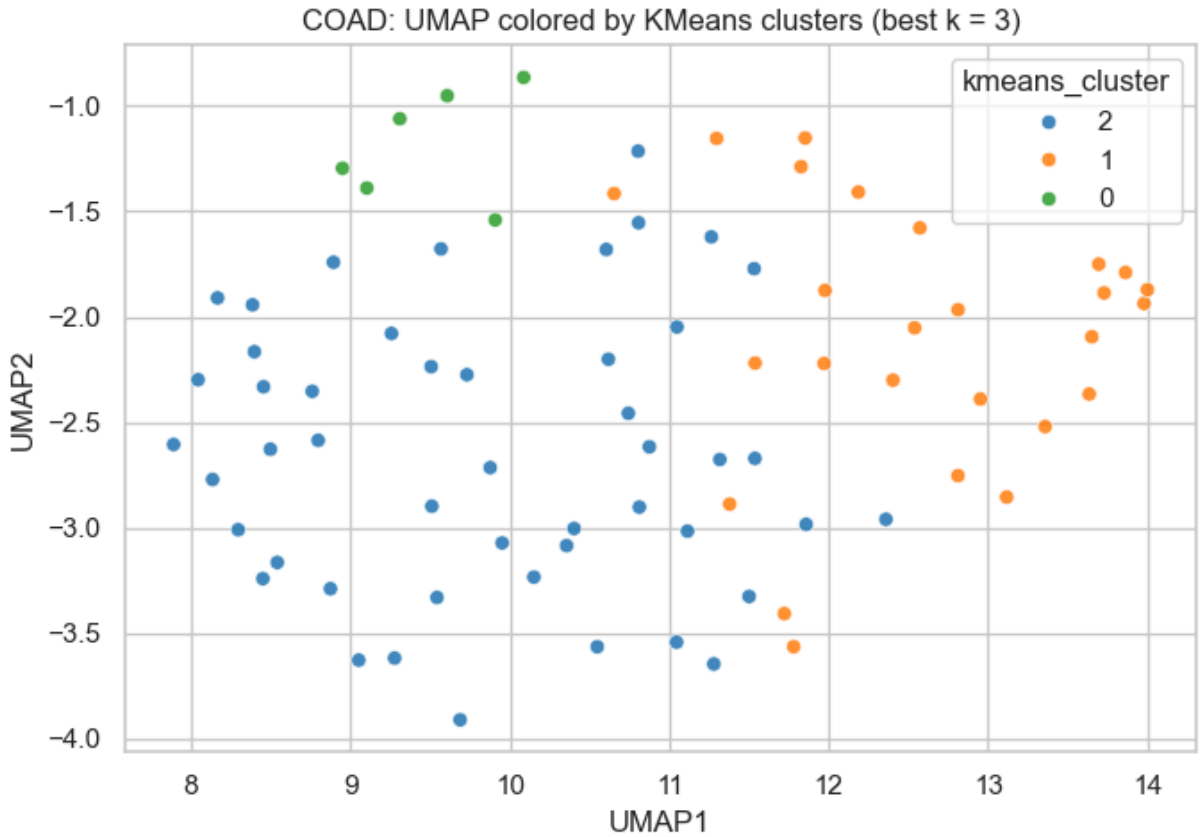

COAD: Hallmark scores by simplified tumor stage
Mean hallmark expression score
5.50
5.25
5.00
4.75
4.50
4.25
4.00
3.75
hallmark_score_type
proliferative_score
apoptosis_resistance_score
Stage II
Stage III
Stage I
Stage IV
Simplified tumor stage


COAD: PCA of 40 hallmark genes
stage_simple
Stage II
Stage III
Stage I
Stage IV
PC2
PC1
2
0
−2
−4
−6
−10
−8
−6
−4
−2
0
2
4


COAD: PCA colored by KMeans clusters (best k = 3)
kmeans_cluster
2
1
0
PC2
PC1
2
0
−2
−4
−6
−10
−5
0
5
10


COAD: KMeans silhouette score by k
best k = 3
Silhouette score
0.15
0.14
0.13
0.12
0.11
0.10
0.09
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Number of clusters (k)


COAD: UMAP of 40 hallmark genes
UMAP2
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
8
9
10
11
12
13
14
UMAP1
stage_simple
Stage II
Stage III
Stage I
Stage IV




```
EDA complete. Results were saved to:
E:\OneDrive - University of Virginia\Will M\Undergraduate Research Courses-Grants\05
_Spring_2026\Computational_Biomedical_BME_2315\Module-4-Cancer\results
```

### 3. Display the global training-set summary

These outputs summarize our overall training expression matrix before subsetting to COAD. The first table reports the total number of genes and samples as well as the global minimum, maximum, mean, and median log2(TPM+1) expression values. The second table lists the genes with the highest average expression across the entire training set, which helps us understand the overall scale and composition of the dataset before narrowing to our hallmark-focused analysis.

In [3]:
```
# After running the EDA workflow, we can access all the major tables and results fr
display(eda_outputs["full_summary"])

# For example, we can display the top mean expression table which shows the top 10
display(eda_outputs["top_mean_expression"])
```

| | n_genes | n_samples | global_min_log2TPMplus1 | global_max_log2TPMplus1 | global_mean_lo |
|---|---|---|---|---|---|
| **0** | 15716 | 1802 | 0.000000 | 17.410280 | |

| | mean_expression |
|---|---|
| **ACTB** | 12.594034 |
| **FTL** | 12.525603 |
| **TPT1** | 12.476975 |
| **B2M** | 12.347839 |
| **GAPDH** | 12.160252 |
| **TMSB10** | 11.998153 |
| **RPS11** | 11.933026 |
| **ACTG1** | 11.866736 |
| **RPS6** | 11.836285 |
| **RPL8** | 11.655261 |
| **RPLP1** | 11.627998 |
| **RPL30** | 11.564944 |
| **RPL37A** | 11.405454 |
| **RPL18** | 11.291977 |
| **RPL11** | 11.282330 |
| **RPLP2** | 11.274173 |
| **GNB2L1** | 11.264597 |
| **RPS8** | 11.264118 |
| **RPS4X** | 11.195749 |
| **RPS16** | 11.060553 |

### 4. Examine metadata completeness and the cleaned COAD metadata

Because metadata completeness varies across cancer types in this teaching dataset, we explicitly examined missingness before moving forward. We also displayed the cleaned COAD metadata to confirm that age was converted to numeric where possible and that the detailed AJCC labels were successfully simplified into broader stage groups. This cleaning step is important because it prepares a more interpretable stage variable for plotting now and for supervised learning later.

In [4]:
```
# We can also display the missingness table which shows the percentage of missing v
# also the cleaned cancer metadata with the new stage columns we created.
display(eda_outputs["missingness"].head(20))

# Display the first 10 rows of the cleaned cancer metadata with the new stage colum
display(
```

```
    eda_outputs["cancer_metadata"][
        ["age_at_diagnosis", "stage_clean", "stage_simple", "stage_binary"]
    ].head(10)
)
```

| | n_non_missing | n_missing | percent_non_missing | percent_missi |
|---|---|---|---|---|
| **cancer_type** | 1802 | 0 | 100.000000 | 0.0000 |
| **history_other_malignancy** | 1730 | 72 | 96.004440 | 3.9955 |
| **gender** | 1730 | 72 | 96.004440 | 3.9955 |
| **race** | 1730 | 72 | 96.004440 | 3.9955 |
| **ethnicity** | 1730 | 72 | 96.004440 | 3.9955 |
| **age_at_diagnosis** | 1730 | 72 | 96.004440 | 3.9955 |
| **bcr_patient_barcode** | 1730 | 72 | 96.004440 | 3.9955 |
| **OS** | 1727 | 75 | 95.837958 | 4.1620 |
| **OS.time** | 1719 | 83 | 95.394007 | 4.6059 |
| **tumor_status** | 1650 | 152 | 91.564928 | 8.4350 |
| **PFI** | 1647 | 155 | 91.398446 | 8.6015 |
| **PFI.time** | 1642 | 160 | 91.120977 | 8.8790 |
| **DSS.time** | 1640 | 162 | 91.009989 | 8.9900 |
| **DSS** | 1625 | 177 | 90.177580 | 9.8224 |
| **histologic_diagnosis** | 1272 | 530 | 70.588235 | 29.4117 |
| **ajcc_pathologic_tumor_stage** | 1144 | 658 | 63.485017 | 36.5149 |
| **ajcc_tumor_pathologic_pt** | 1142 | 660 | 63.374029 | 36.6259 |
| **ajcc_metastasis_pathologic_pm** | 1142 | 660 | 63.374029 | 36.6259 |
| **ajcc_nodes_pathologic_pn** | 1142 | 660 | 63.374029 | 36.6259 |
| **DFI** | 813 | 989 | 45.116537 | 54.8834 |

| sample | age_at_diagnosis | stage_clean | stage_simple | stage_binary |
|---|---|---|---|---|
| **TCGA-AA-3532-01A-01R-0821-07** | 63.000000 | Stage IIA | Stage II | Early |
| **TCGA-AA-3842-01A-01R-1022-07** | 51.000000 | Stage IIIA | Stage III | Late |
| **TCGA-G4-6310-01A-11R-1723-07** | 69.000000 | Stage IIIB | Stage III | Late |
| **TCGA-AA-3975-01A-01R-1022-07** | 80.000000 | Stage I | Stage I | Early |
| **TCGA-AM-5820-01A-01R-1653-07** | 59.000000 | Stage IVA | Stage IV | Late |
| **TCGA-AA-A02W-01A-01R-A00A-07** | 73.000000 | Stage I | Stage I | Early |
| **TCGA-CA-6719-01A-11R-1839-07** | 77.000000 | Stage IIA | Stage II | Early |
| **TCGA-G4-6295-01A-11R-1723-07** | 70.000000 | Stage II | Stage II | Early |
| **TCGA-AA-3949-01A-01R-1022-07** | 87.000000 | Stage IIIB | Stage III | Late |
| **TCGA-AA-3815-01A-01R-1022-07** | 65.000000 | Stage IIA | Stage II | Early |

## 5. Confirm that the selected hallmark genes are present in the COAD subset

Before interpreting any downstream result, we verified that the requested hallmark genes were actually present in the local expression matrix. The presence table confirms which genes were found, and the summary-statistics table provides the mean, median, and variance of each hallmark gene across the COAD samples. This step ensures that the selected features are available and gives a first look at which genes are relatively stable or variable within this cohort.

In [5]:
```
# Finally, we can also display the hallmark gene presence table which shows the pre
# Also the summary statistics for these genes.
display(eda_outputs["hallmark_gene_presence"])
display(eda_outputs["hallmark_gene_stats"])
```

| | hallmark | gene | present_in_dataset |
|---|---|---|---|
| **0** | Sustained_Proliferative_Signaling | EGFR | True |
| **1** | Sustained_Proliferative_Signaling | ERBB2 | True |
| **2** | Sustained_Proliferative_Signaling | KRAS | True |
| **3** | Sustained_Proliferative_Signaling | NRAS | True |
| **4** | Sustained_Proliferative_Signaling | BRAF | True |
| **5** | Sustained_Proliferative_Signaling | PIK3CA | True |
| **6** | Sustained_Proliferative_Signaling | AKT1 | True |
| **7** | Sustained_Proliferative_Signaling | MYC | True |
| **8** | Sustained_Proliferative_Signaling | FOS | True |
| **9** | Sustained_Proliferative_Signaling | JUN | True |
| **10** | Sustained_Proliferative_Signaling | MET | True |
| **11** | Sustained_Proliferative_Signaling | IGF1R | True |
| **12** | Sustained_Proliferative_Signaling | FGFR1 | True |
| **13** | Sustained_Proliferative_Signaling | FGFR2 | True |
| **14** | Sustained_Proliferative_Signaling | EGF | True |
| **15** | Sustained_Proliferative_Signaling | HGF | True |
| **16** | Sustained_Proliferative_Signaling | SOS1 | True |
| **17** | Sustained_Proliferative_Signaling | GRB2 | True |
| **18** | Sustained_Proliferative_Signaling | MAPK1 | True |
| **19** | Sustained_Proliferative_Signaling | MAPK3 | True |
| **20** | Resist_Apoptosis | TP53 | True |
| **21** | Resist_Apoptosis | BCL2 | True |
| **22** | Resist_Apoptosis | BCL2L1 | True |
| **23** | Resist_Apoptosis | MCL1 | True |
| **24** | Resist_Apoptosis | CASP8 | True |
| **25** | Resist_Apoptosis | CASP3 | True |
| **26** | Resist_Apoptosis | BAX | True |
| **27** | Resist_Apoptosis | BAK1 | True |
| **28** | Resist_Apoptosis | APAF1 | True |
| **29** | Resist_Apoptosis | BID | True |

| | hallmark | gene | present_in_dataset |
|---|---|---|---|
| **30** | Resist_Apoptosis | CASP9 | True |
| **31** | Resist_Apoptosis | FAS | True |
| **32** | Resist_Apoptosis | FASLG | True |
| **33** | Resist_Apoptosis | BAD | True |
| **34** | Resist_Apoptosis | BBC3 | True |
| **35** | Resist_Apoptosis | BCL2L11 | True |
| **36** | Resist_Apoptosis | CFLAR | True |
| **37** | Resist_Apoptosis | DIABLO | True |
| **38** | Resist_Apoptosis | TNF | True |
| **39** | Resist_Apoptosis | TNFRSF10B | True |

| | mean_expression | median_expression | variance_across_samples |
|---|---|---|---|
| **MYC** | 8.228511 | 8.221083 | 0.659806 |
| **FOS** | 7.957519 | 7.856431 | 1.629929 |
| **JUN** | 7.618991 | 7.732972 | 0.609185 |
| **MCL1** | 7.431396 | 7.501133 | 0.490321 |
| **BCL2L1** | 7.398317 | 7.387813 | 0.527915 |
| **MAPK3** | 7.251998 | 7.294059 | 0.606089 |
| **BAX** | 7.229717 | 7.268432 | 0.492506 |
| **AKT1** | 7.080532 | 7.166449 | 0.234295 |
| **GRB2** | 6.810953 | 6.816642 | 0.198127 |
| **ERBB2** | 6.782761 | 6.903818 | 0.675691 |
| **TP53** | 6.080440 | 6.598347 | 1.795712 |
| **BAD** | 6.038458 | 6.123912 | 0.308288 |
| **DIABLO** | 5.965331 | 5.964623 | 0.111331 |
| **MET** | 5.942898 | 5.879211 | 0.717186 |
| **BID** | 5.829742 | 5.781805 | 0.251465 |
| **MAPK1** | 5.705211 | 5.766001 | 0.310708 |
| **BAK1** | 5.660182 | 5.656553 | 0.338468 |
| **TNFRSF10B** | 5.576977 | 5.615052 | 0.548658 |
| **CASP3** | 5.522967 | 5.678818 | 0.371093 |
| **NRAS** | 5.477788 | 5.670852 | 0.745258 |
| **BBC3** | 4.617883 | 4.658929 | 0.410255 |
| **KRAS** | 4.602230 | 4.672501 | 0.763439 |
| **CASP8** | 4.347573 | 4.397027 | 0.343758 |
| **CFLAR** | 3.926365 | 3.904053 | 0.406354 |
| **SOS1** | 3.791937 | 3.822083 | 0.534900 |
| **CASP9** | 3.782139 | 3.823212 | 0.258526 |
| **IGF1R** | 3.769224 | 3.701273 | 0.551881 |
| **FAS** | 3.736817 | 3.694270 | 1.060056 |
| **EGFR** | 3.470916 | 3.455831 | 0.627900 |
| **APAF1** | 3.358721 | 3.413749 | 0.507264 |

| | mean_expression | median_expression | variance_across_samples |
|---|---|---|---|
| **FGFR1** | 3.284186 | 3.192755 | 1.169687 |
| **BCL2L11** | 3.273790 | 3.326452 | 0.400095 |
| **PIK3CA** | 3.229337 | 3.308622 | 0.627246 |
| **FGFR2** | 2.575545 | 2.484944 | 1.273997 |
| **BRAF** | 2.216751 | 2.186404 | 0.426419 |
| **BCL2** | 1.938258 | 1.990227 | 0.795100 |
| **FASLG** | 0.970790 | 0.775964 | 0.546242 |
| **HGF** | 0.968300 | 0.855008 | 0.324848 |
| **EGF** | 0.881202 | 0.338483 | 1.149523 |
| **TNF** | 0.295415 | 0.247102 | 0.047064 |

## 6. Review the sample-level hallmark-score table

We next computed one simple score per hallmark by averaging expression across the genes in each hallmark set for each sample. These sample-level hallmark scores give us a compact way to compare the two biological programs across tumors and provide a natural bridge between gene-level data and stage-based comparisons.

In [6]:
```
# We can also display the merged scores table which shows the proliferative score a
display(
    eda_outputs["merged_scores"][
        ["proliferative_score", "apoptosis_resistance_score", "stage_simple", "stag
    ].head(10)
)
```

| | proliferative_score | apoptosis_resistance_score | stage_simple | stage_binary |
|---|---|---|---|---|
| **TCGA-AA-3532-01A-01R-0821-07** | 4.805219 | 4.681239 | Stage II | Early |
| **TCGA-AA-3842-01A-01R-1022-07** | 4.869162 | 4.530510 | Stage III | Late |
| **TCGA-G4-6310-01A-11R-1723-07** | 5.144955 | 4.636565 | Stage III | Late |
| **TCGA-AA-3975-01A-01R-1022-07** | 4.991320 | 4.681024 | Stage I | Early |
| **TCGA-AM-5820-01A-01R-1653-07** | 5.280900 | 4.767320 | Stage IV | Late |
| **TCGA-AA-A02W-01A-01R-A00A-07** | 5.101496 | 4.214700 | Stage I | Early |
| **TCGA-CA-6719-01A-11R-1839-07** | 5.597140 | 4.554142 | Stage II | Early |
| **TCGA-G4-6295-01A-11R-1723-07** | 4.938214 | 4.760688 | Stage II | Early |
| **TCGA-AA-3949-01A-01R-1022-07** | 4.310490 | 4.755428 | Stage III | Late |
| **TCGA-AA-3815-01A-01R-1022-07** | 4.827595 | 5.002941 | Stage II | Early |

## 7. Inspect the PCA coordinates used for dimensionality reduction

This table contains the PCA coordinates for each COAD sample after dimensionality reduction of the 40-gene hallmark matrix. PCA allows us to visualize the dominant axes of variation in the selected gene set and check whether samples appear to separate by tumor stage or by other structure present in the data.

In [7]:
```
# Finally, we can also display the PCA table which shows the first 10 rows of the P
display(eda_outputs["pca_df"].head(10))
```

| | PC1 | PC2 | proliferative_score | apoptosis_resistance_score | cancer_type | bcr_ |
|---|---|---|---|---|---|---|
| **TCGA-AA-3532-01A-01R-0821-07** | 0.375019 | -0.732955 | 4.805219 | 4.681239 | COAD | |
| **TCGA-AA-3842-01A-01R-1022-07** | -0.419658 | -0.675840 | 4.869162 | 4.530510 | COAD | |
| **TCGA-G4-6310-01A-11R-1723-07** | 1.418868 | 1.345956 | 5.144955 | 4.636565 | COAD | |
| **TCGA-AA-3975-01A-01R-1022-07** | -0.624553 | 1.709509 | 4.991320 | 4.681024 | COAD | |
| **TCGA-AM-5820-01A-01R-1653-07** | 1.369316 | 0.365577 | 5.280900 | 4.767320 | COAD | |
| **TCGA-AA-A02W-01A-01R-A00A-07** | 0.548001 | -0.850722 | 5.101496 | 4.214700 | COAD | |
| **TCGA-CA-6719-01A-11R-1839-07** | 2.829912 | 1.784933 | 5.597140 | 4.554142 | COAD | |

| | PC1 | PC2 | proliferative_score | apoptosis_resistance_score | cancer_type | bcr_ |
|---|---|---|---|---|---|---|
| **TCGA-G4-6295-01A-11R-1723-07** | 1.635818 | 1.189609 | 4.938214 | 4.760688 | COAD | |
| **TCGA-AA-3949-01A-01R-1022-07** | -0.125615 | -3.696958 | 4.310490 | 4.755428 | COAD | |
| **TCGA-AA-3815-01A-01R-1022-07** | -0.451677 | 0.444962 | 4.827595 | 5.002941 | COAD | |

10 rows × 37 columns

## 8. Inspect the K-means cluster evaluation table

We used K-means clustering as an exploratory grouping method and compared candidate values of $k$ using silhouette score. This table summarizes the clustering results for each tested value of $k$, including inertia and silhouette score, and helps identify the most reasonable first-pass number of clusters for the COAD hallmark-expression data.

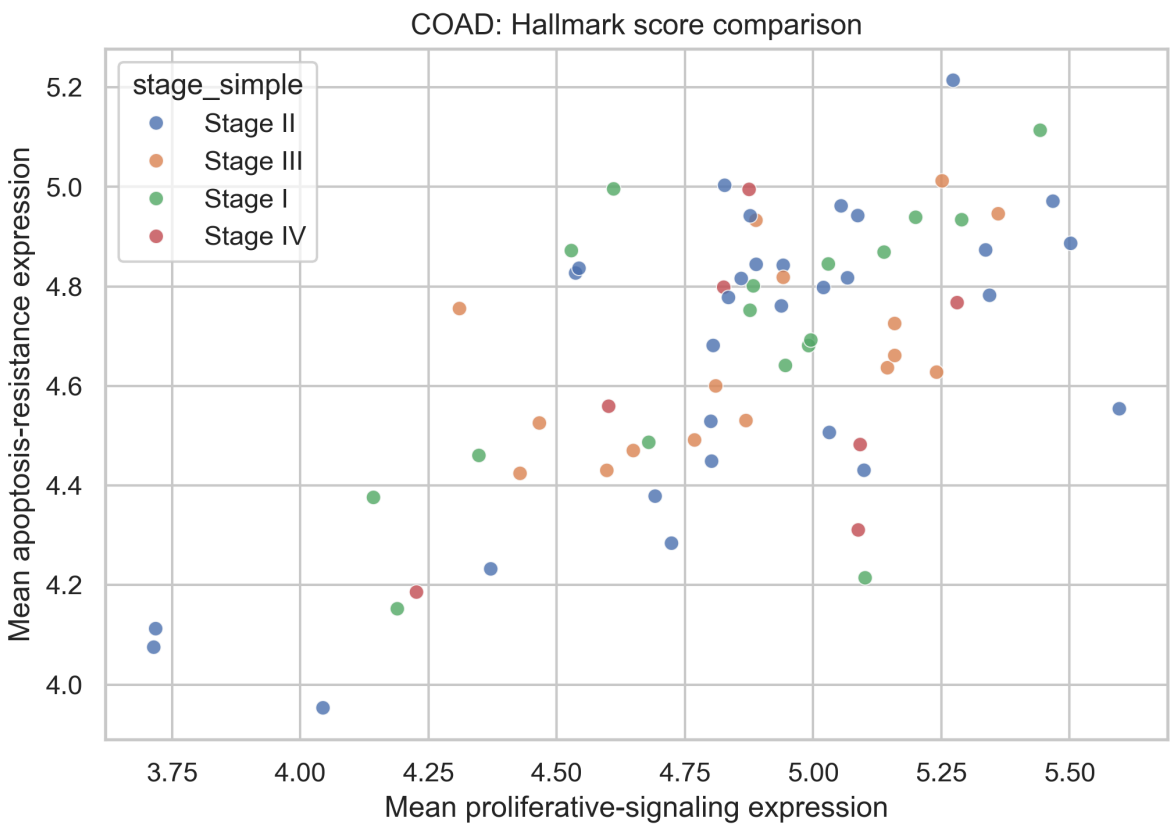
In [8]:
```
# We can also display the cluster-evaluation table, which shows the inertia and sil
display(eda_outputs["cluster_eval"])
```

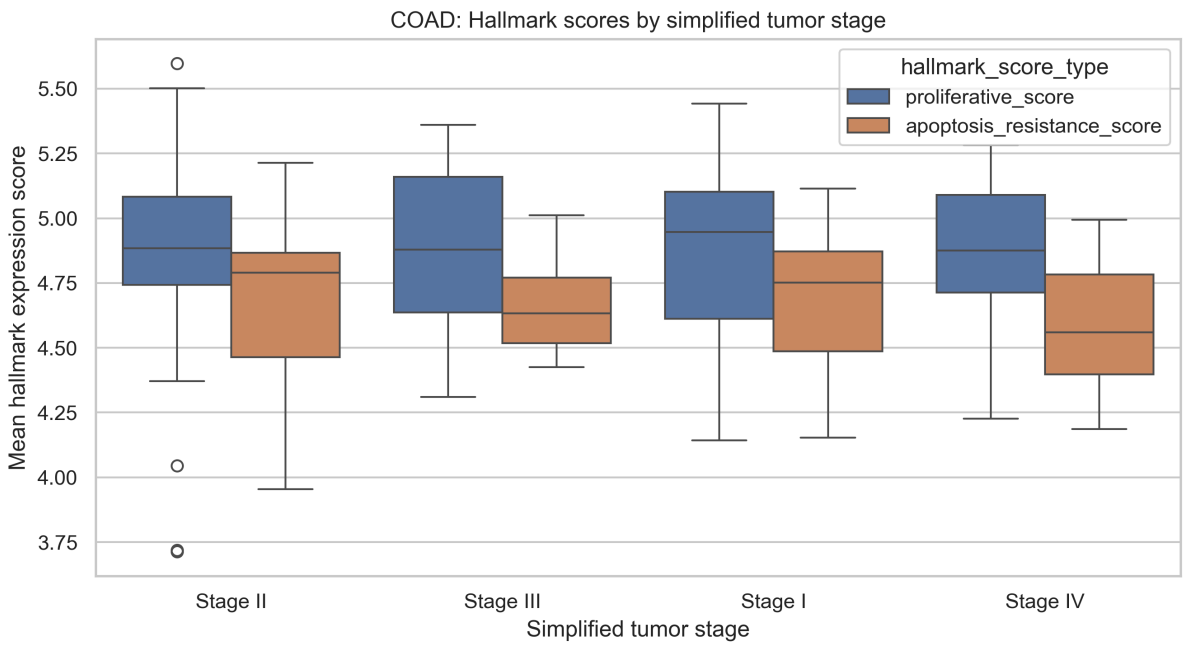
| | k | inertia | silhouette_score |
|---|---|---|---|
| **0** | 2 | 2682.814190 | 0.150278 |
| **1** | 3 | 2411.957810 | 0.153595 |
| **2** | 4 | 2189.351156 | 0.103196 |
| **3** | 5 | 2057.638214 | 0.093785 |
| **4** | 6 | 1957.743396 | 0.086763 |

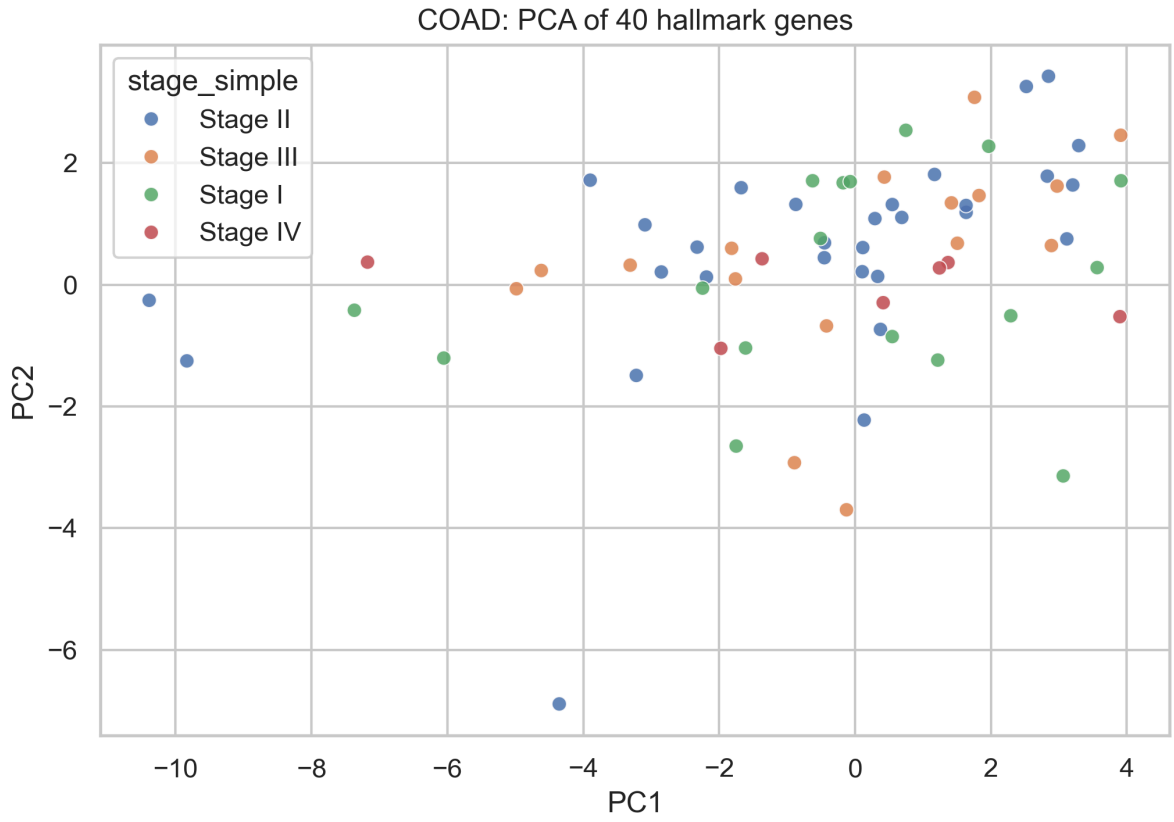
## 9. Review the saved output figures

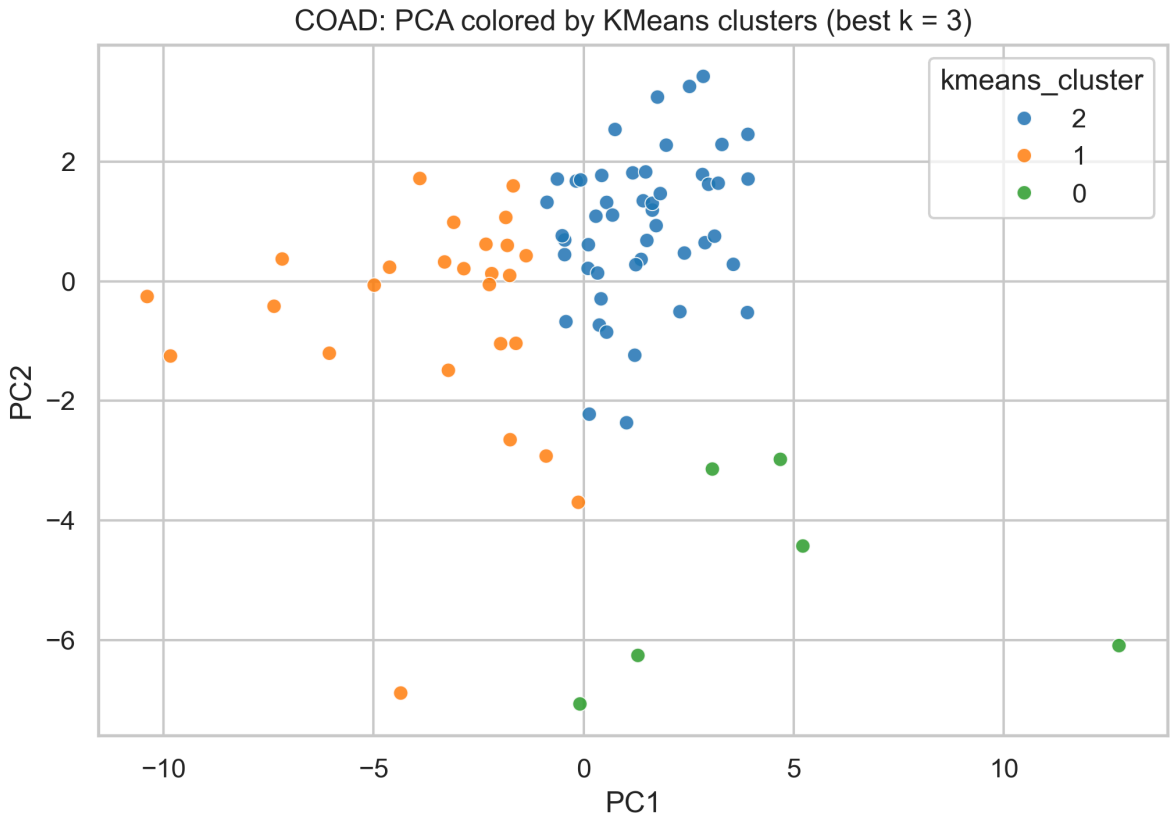
These figures summarize our current exploratory analysis. The hallmark-score scatterplot compares the two hallmark programs directly across samples, the stage-based boxplot asks whether those scores vary across simplified tumor stages, and the PCA and clustering plots visualize broader structure within the 40-gene hallmark panel. The silhouette plot helps justify the current clustering choice by showing how cluster quality changes with the number of clusters.

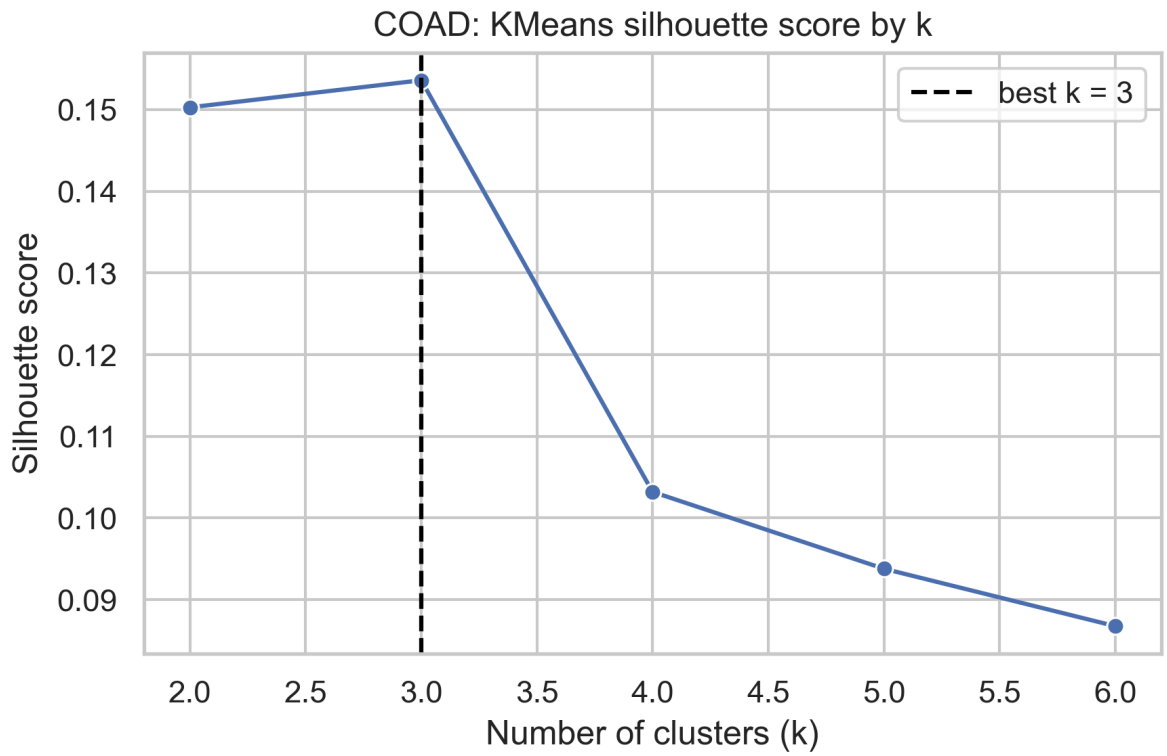
In [9]:
```
# Finally, we can also display the major figures we generated during our EDA workfl
from IPython.display import Image, display

results_dir = PROJECT_ROOT / "results"

display(Image(filename=str(results_dir / "COAD_hallmark_score_scatter.png")))
display(Image(filename=str(results_dir / "COAD_hallmark_scores_by_stage.png")))
display(Image(filename=str(results_dir / "COAD_PCA_by_stage.png")))
display(Image(filename=str(results_dir / "COAD_PCA_by_cluster.png")))
display(Image(filename=str(results_dir / "COAD_kmeans_silhouette.png")))
```




COAD: Hallmark scores by simplified tumor stage
hallmark_score_type
proliferative_score
apoptosis_resistance_score
Mean hallmark expression score
5.50
5.25
5.00
4.75
4.50
4.25
4.00
3.75
Stage II
Stage III
Stage I
Stage IV
Simplified tumor stage


COAD: PCA of 40 hallmark genes
stage_simple
Stage II
Stage III
Stage I
Stage IV
PC2
2
0
−2
−4
−6
−10
−8
−6
−4
−2
0
2
4
PC1




## 10. Display the UMAP coordinates

UMAP provides a nonlinear view of local structure in the same hallmark-gene matrix. We use it here as a complementary visualization to PCA rather than as a final conclusion. Comparing

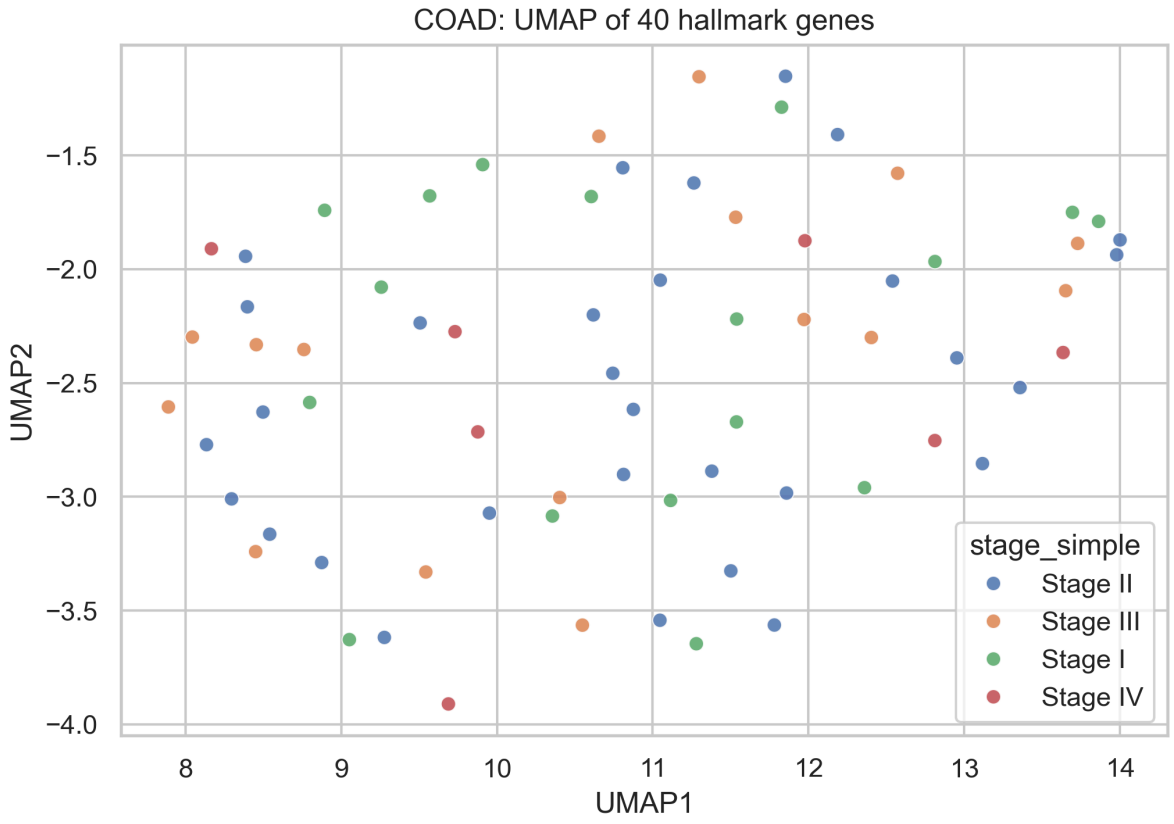
PCA and UMAP helps us judge whether the apparent structure is stable across different dimensionality-reduction approaches.

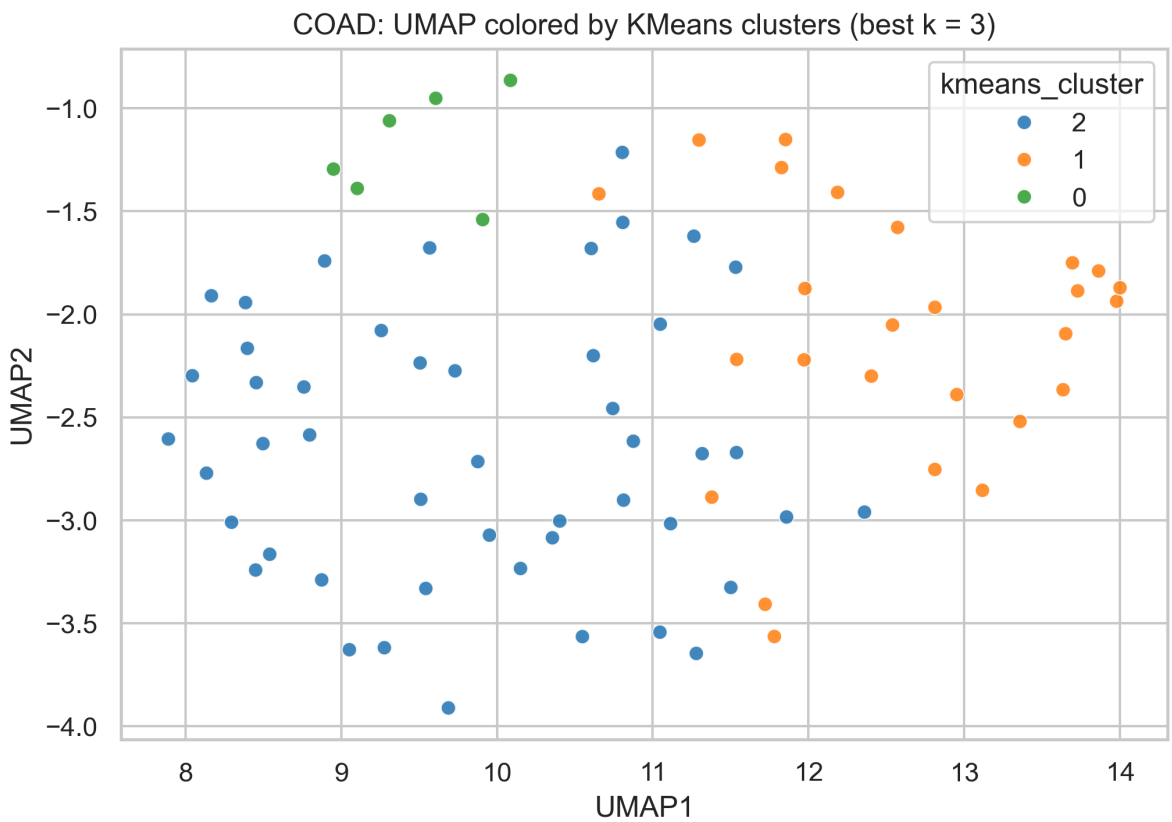
In [10]:
```
# If UMAP was available and dovnloaded in our current environment, we can also disp
if eda_outputs["umap_df"] is not None:
    display(eda_outputs["umap_df"].head(10))
    display(Image(filename=str(results_dir / "COAD_UMAP_by_stage.png")))
    display(Image(filename=str(results_dir / "COAD_UMAP_by_cluster.png")))
else:
    print("No UMAP available.")
```

| | UMAP1 | UMAP2 | proliferative_score | apoptosis_resistance_score | cancer_type | bcr |
|---|---|---|---|---|---|---|
| **TCGA-AA-3532-01A-01R-0821-07** | 11.264119 | -1.620861 | 4.805219 | 4.681239 | COAD | |
| **TCGA-AA-3842-01A-01R-1022-07** | 11.532536 | -1.771479 | 4.869162 | 4.530510 | COAD | |
| **TCGA-G4-6310-01A-11R-1723-07** | 8.761057 | -2.352844 | 5.144955 | 4.636565 | COAD | |
| **TCGA-AA-3975-01A-01R-1022-07** | 12.359118 | -2.960093 | 4.991320 | 4.681024 | COAD | |
| **TCGA-AM-5820-01A-01R-1653-07** | 9.730977 | -2.274360 | 5.280900 | 4.767320 | COAD | |
| **TCGA-AA-A02W-01A-01R-A00A-07** | 9.052812 | -3.627867 | 5.101496 | 4.214700 | COAD | |
| **TCGA-CA-6719-01A-11R-1839-07** | 8.296686 | -3.009314 | 5.597140 | 4.554142 | COAD | |

| | UMAP1 | UMAP2 | proliferative_score | apoptosis_resistance_score | cancer_type | bcr |
|---|---|---|---|---|---|---|
| **TCGA-G4-6295-01A-11R-1723-07** | 8.875164 | -3.289126 | 4.938214 | 4.760688 | COAD | |
| **TCGA-AA-3949-01A-01R-1022-07** | 10.654593 | -1.415373 | 4.310490 | 4.755428 | COAD | |
| **TCGA-AA-3815-01A-01R-1022-07** | 11.048312 | -2.047903 | 4.827595 | 5.002941 | COAD | |

10 rows × 37 columns





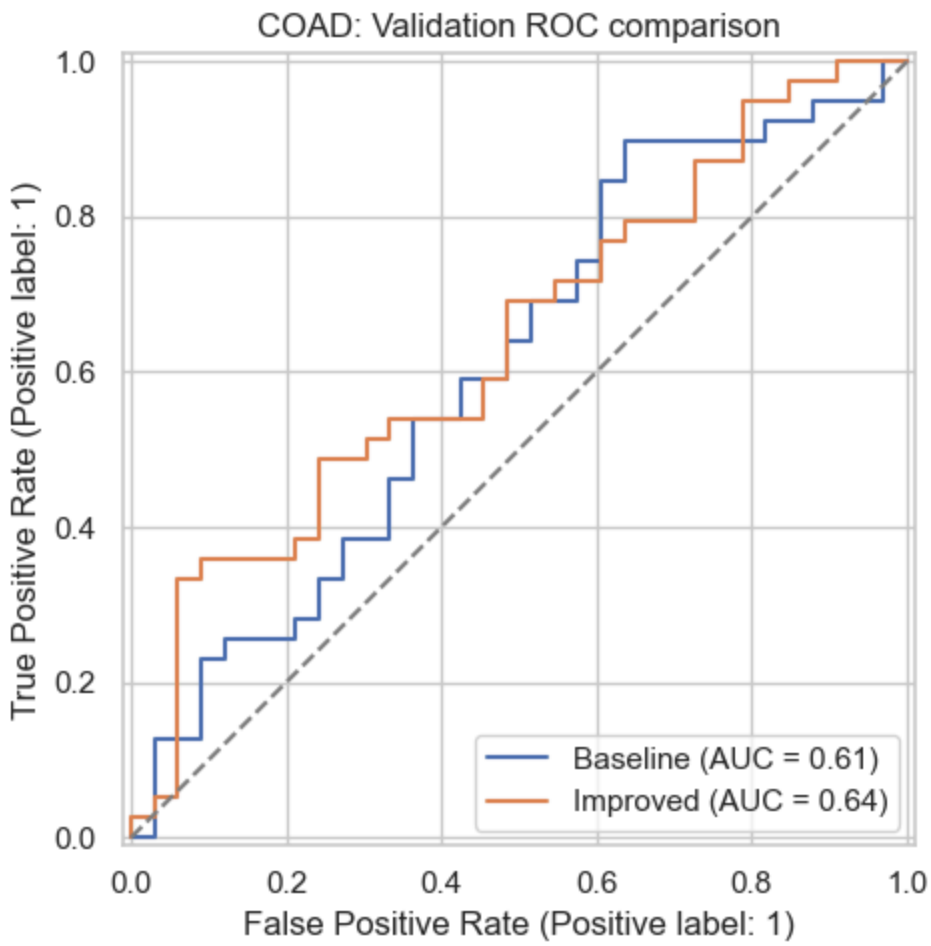

## Overall Unsupervised Patterns Interpretation

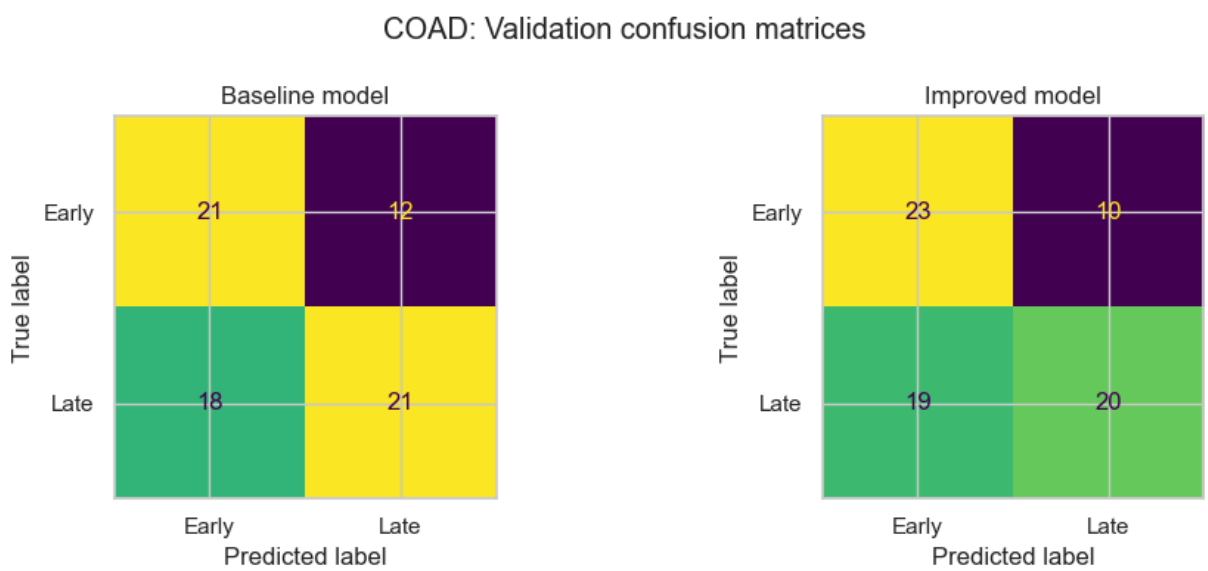
Our unsupervised results suggest that the two hallmark programs are related but do not clearly separate tumor stage. In our hallmark-score scatterplot, samples with **higher proliferative scores** generally also tended to have **higher apoptosis-resistance scores**, indicating partial coordination between the two hallmarks. However, the stage-based boxplots clearly show substantial overlap across Stage I–IV, suggesting that these summary scores alone **do not cleanly distinguish stage**. On a similar note, PCA and UMAP did not reveal strong stage-separated groupings, and the best K-means solution had a low silhouette score of 0.154 for k = 3, indicating there was a weak cluster structure in our 40-gene feature space. Because the PCA, UMAP, and clustering analyses showed only weak separation by stage, we expected the supervised classifier to capture at most modest stage-related signal from this 40-gene hallmark panel.

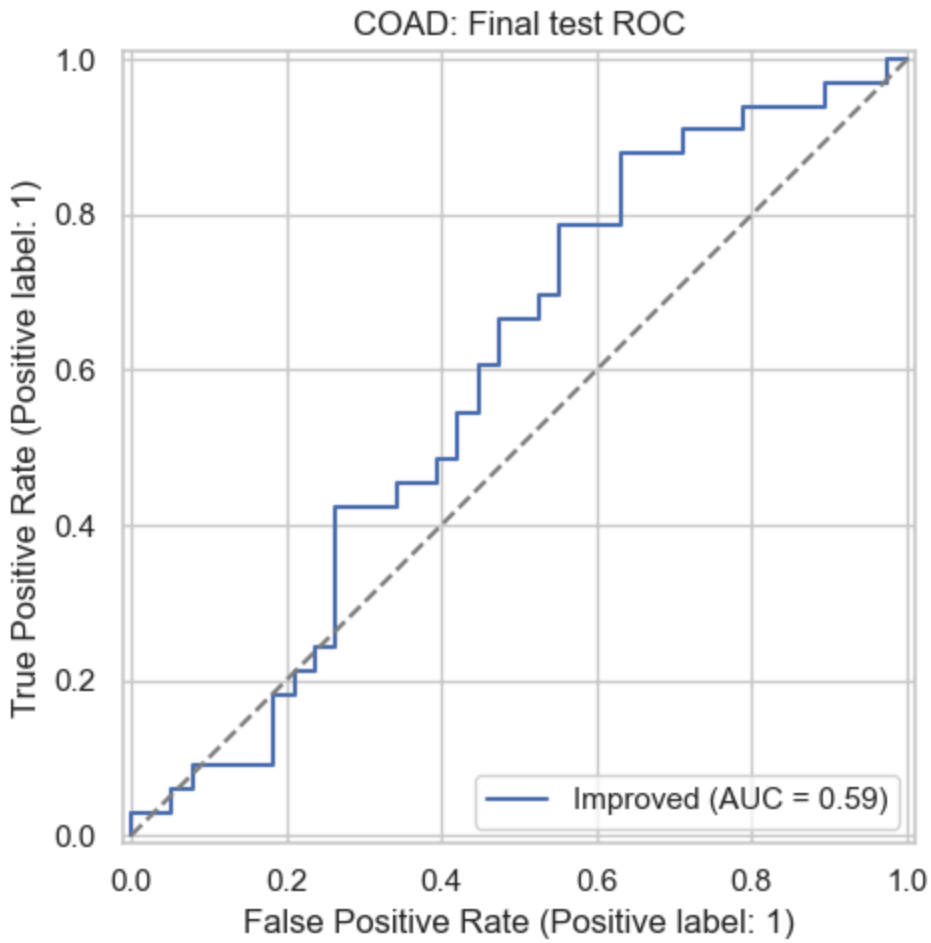
## 11. Run the supervised training/validation workflow

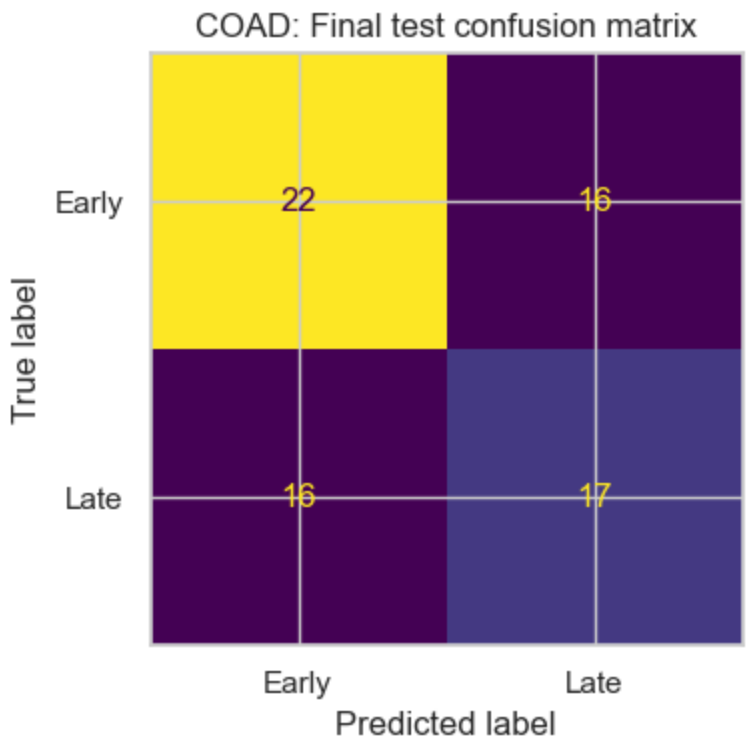
In this step, we transition from exploratory analysis to supervised learning. Our goal is now to predict whether a COAD tumor belongs to the **Early** or **Late** stage group using hallmark-expression features. Following the workflow discussed in class, we use the **training set** to fit candidate models, the **validation set** to compare and select the final model, and the **test set** only once at the end to estimate final out-of-sample performance.

In [11]:
```
# After completing the EDA, we now move to supervised learning.
# This fits candidate models on the training set, compares them on the validation s
# selects the final model, and then evaluates that selected model once on the held-
supervised_results = main.run_supervised_modeling()
```






**12. Compare baseline and improved model metrics**

This table compares the baseline and improved logistic-regression models on both the training and validation datasets. Reporting both in-sample and out-of-sample metrics helps us determine whether the model is learning a useful biological pattern that generalizes, rather than simply memorizing the training data.

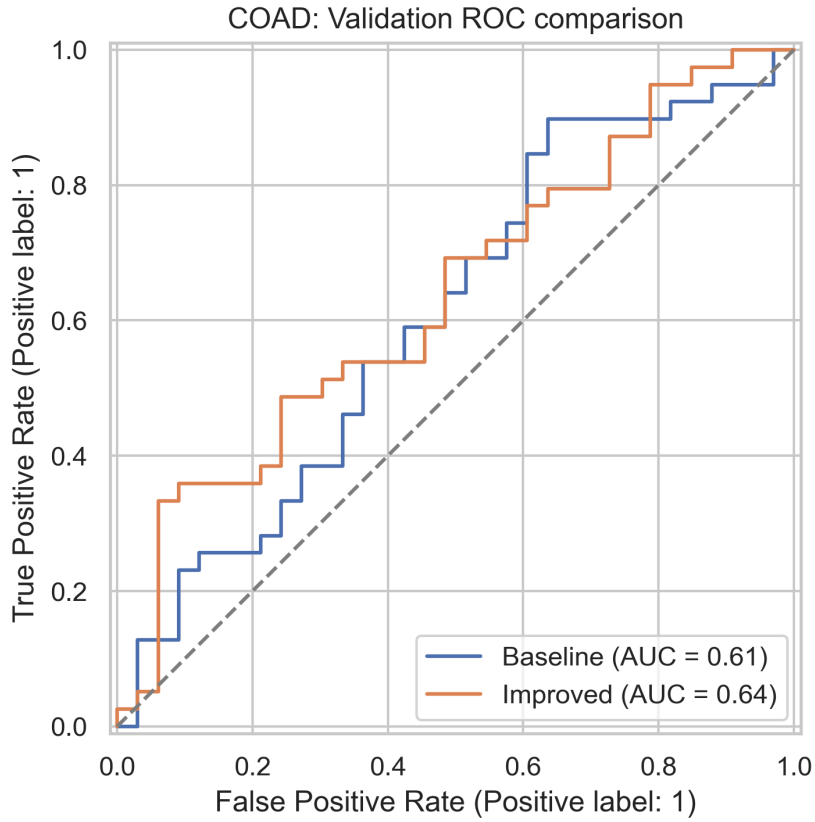
In [12]:
```
# We can also display the ROC curve comparison data for the validation set
# compares the baseline and improved models.
display(supervised_results["comparison_df"])
```

| | model | split | n_samples | accuracy | balanced |
|---|---|---|---|---|---|
| **0** | baseline_logreg_40genes | train | 70 | 0.900000 | |
| **1** | baseline_logreg_40genes | validation | 72 | 0.583333 | |
| **2** | improved_logreg_40genes_plus_scores_age_balanced | train | 70 | 0.857143 | |
| **3** | improved_logreg_40genes_plus_scores_age_balanced | validation | 72 | 0.597222 | |

## 13. Display the validation ROC comparison

The ROC curves summarize how well each model separates Early-stage from Late-stage tumors across all possible decision thresholds. Because logistic regression outputs probability scores, AUROC is a useful threshold-independent metric for comparing the baseline and improved models.

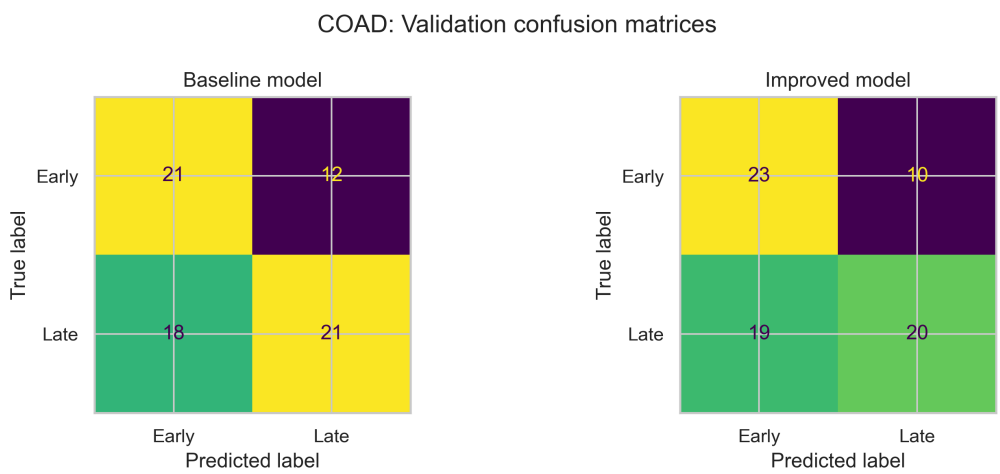
In [13]:
```
# Finally, we can also display the ROC curve comparison figure for the validation s
display(Image(filename=str(results_dir / "COAD_validation_ROC_comparison.png")))
```




## 14. Display the validation confusion matrices

The confusion matrices show model behavior at the default decision threshold of 0.5. They make it easier to see the balance between correctly identifying Late-stage tumors and incorrectly labeling Early-stage tumors as Late, which is important for interpreting model error in a clinically meaningful way.

In [14]:
```
# We can also display the confusion matrix comparison figure for the validation set
display(Image(filename=str(results_dir / "COAD_validation_confusion_matrices.png")))
```




## 15. Top coefficients from the improved model

These coefficients provide a first-pass look at which features most strongly influence the improved logistic-regression model. Because logistic regression is interpretable, the coefficient table helps connect predictive performance back to the biological story of proliferative signaling, apoptosis resistance, and tumor progression.

In [15]:
```
#  display the coefficient table for the improved model which shows the top feature
display(supervised_results["improved"]["coef_df"].head(15))
```

| | feature | coefficient | abs_coefficient |
|---|---|---|---|
| **0** | EGFR | 0.336864 | 0.336864 |
| **12** | FGFR1 | 0.287371 | 0.287371 |
| **30** | CASP9 | -0.278940 | 0.278940 |
| **35** | BCL2L11 | 0.272989 | 0.272989 |
| **7** | MYC | -0.261921 | 0.261921 |
| **1** | ERBB2 | -0.257498 | 0.257498 |
| **9** | JUN | -0.223860 | 0.223860 |
| **26** | BAX | -0.205219 | 0.205219 |
| **31** | FAS | -0.197033 | 0.197033 |
| **10** | MET | -0.194539 | 0.194539 |
| **15** | HGF | -0.177590 | 0.177590 |
| **5** | PIK3CA | 0.177555 | 0.177555 |
| **29** | BID | -0.175142 | 0.175142 |
| **36** | CFLAR | 0.147117 | 0.147117 |
| **38** | TNF | -0.138413 | 0.138413 |

## Supervised Results Interpretation

Our supervised results show that our **improved** logistic-regression model generalized slightly better than our **baseline** model, but performance remained modest overall. On the validation set, the **improved** model achieved **higher AUROC**, balanced accuracy, and accuracy than the **baseline**, which suggests to us that adding hallmark-summary scores, age, class balancing, and stronger regularization was directionally helpful. However, the gap between training and validation performance remained substantial for both of our models, which indicates that the selected hallmark-expression features alone do not provide strong stage-prediction power in our dataset. Because the baseline model performed much better on the training set than on the validation set, we interpreted this pattern as overfitting and designed the improved model to reduce variance and improve generalization.

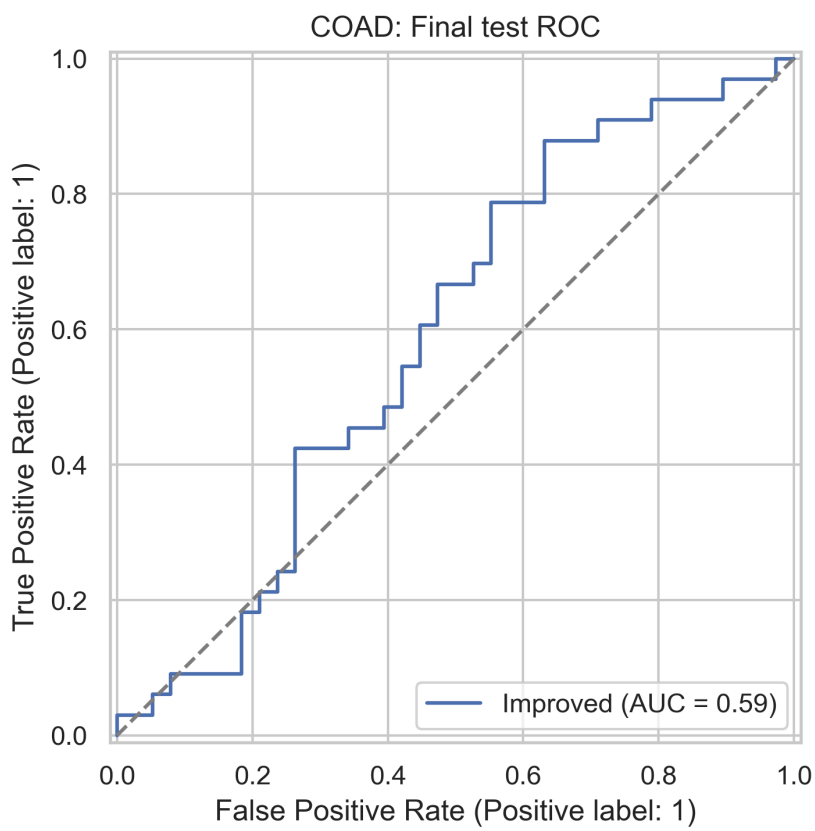
## 16. Evaluate the selected final model on the held-out test set

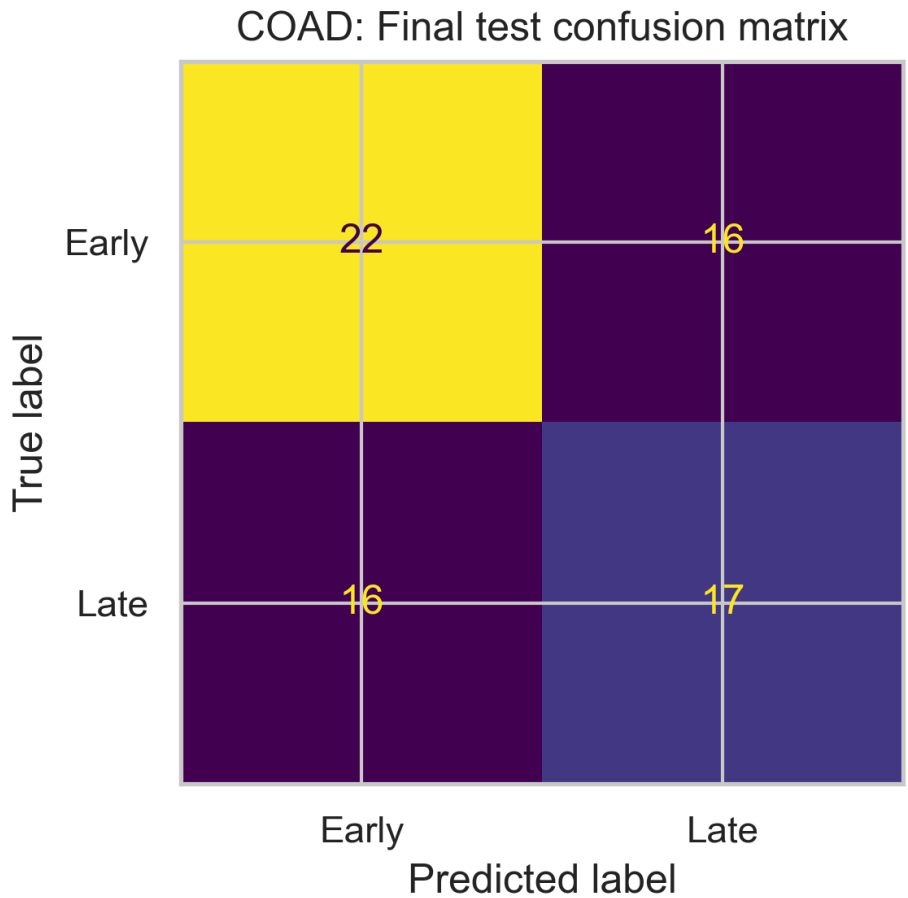
Following the final workflow discussed in class, we used the validation set to compare the baseline and improved models and then applied only the selected final model to the unseen test set. This provides our final estimate of out-of-sample performance on COAD samples that were not used during training or model selection.

In [16]:
```
# display the final test results for the selected model on the held-out test set, i
display(supervised_results["final_test_results"]["test_metrics"])
print("Selected model from validation:", supervised_results["selected_model_name"])

# ROC curve and confusion matrix for the final selected model on the held-out test
display(Image(filename=str(results_dir / "COAD_final_test_ROC.png")))
display(Image(filename=str(results_dir / "COAD_final_test_confusion_matrix.png")))
display(supervised_results["final_metrics_df"])
```

| | model | split | n_samples | accuracy | balanced_accu |
|---|---|---|---|---|---|
| **0** | improved_logreg_40genes_plus_scores_age_balanced | test | 71 | 0.549296 | 0.54 |

```
Selected model from validation: improved
```






| | model | split | n_samples | accuracy | balanced |
|---|---|---|---|---|---|
| **0** | baseline_logreg_40genes | train | 70 | 0.900000 | |
| **1** | baseline_logreg_40genes | validation | 72 | 0.583333 | |
| **2** | improved_logreg_40genes_plus_scores_age_balanced | train | 70 | 0.857143 | |
| **3** | improved_logreg_40genes_plus_scores_age_balanced | validation | 72 | 0.597222 | |
| **4** | improved_logreg_40genes_plus_scores_age_balanced | test | 71 | 0.549296 | |

### Final Test-Set Interpretation

The held-out test set provides our strongest estimate of model generalization because it was not used to fit the model or to choose between candidate models. If the test-set metrics are similar to the validation-set metrics, that suggests our validation estimate was reasonably stable. If the test performance drops noticeably, that would suggest that the model still has limited robustness and remains somewhat sensitive to the specific training/validation split.

In our case, the final selected model achieved **0.589 AUROC**, **0.549 accuracy**, and **0.547 balanced accuracy** on the test set. These values were slightly lower than the validation-set metrics, suggesting that the model's generalization remained limited when applied to

completely unseen data. Overall, the test results indicate that our final model provides only **modest** discrimination between Early- and Late-stage COAD tumors and is not strong enough to support any clinical predictive use.

# Verify and validate your analysis:

We evaluated model performance using both **threshold-independent** and **threshold-dependent** metrics. **AUROC** was our primary threshold-independent metric because logistic regression outputs probabilities, and AUROC summarizes how well the model ranks Late-stage tumors above Early-stage tumors across all thresholds. At the default threshold of **0.5**, we also examined the **confusion matrix**, **accuracy**, **balanced accuracy**, **precision**, **recall**, and **F1 score**.

We first used the **training set** to fit our models and estimate in-sample performance. We then used the **validation set** to compare candidate models and assess out-of-sample generalization. The baseline logistic-regression model performed very well on the training set (**AUROC = 0.974**, **accuracy = 0.900**) but much worse on the validation set (**AUROC = 0.608**, **accuracy = 0.583**), which suggests overfitting. The improved model reduced training-set performance somewhat (**AUROC = 0.917**, **accuracy = 0.857**) but modestly improved validation-set performance (**AUROC = 0.637**, **accuracy = 0.597**, **balanced accuracy = 0.605**), indicating that adding hallmark-summary scores, age at diagnosis, class balancing, and stronger regularization reduced overfitting slightly.

Following the final workflow from class, we then selected the better-performing model based on **validation AUROC** and applied that selected model once to the **held-out test set**. This test-set evaluation provides our final out-of-sample error estimate because the test data were not used during model fitting or model selection. Our final selected model achieved **0.589 AUROC**, **0.549 accuracy**, and **0.547 balanced accuracy** on the test set. The test confusion matrix showed 22 correctly classified Early-stage tumors and 17 correctly classified Late-stage tumors, with 16 misclassifications in each error direction, further emphasizing that the classifier's discrimination was limited.

Overall, our model shows **modest predictive ability** rather than strong stage discrimination. The gap between training performance and validation/test performance indicates that the selected hallmark-expression features capture some biologically meaningful signal related to stage, but not enough to produce a robust stage classifier on their own.

# Conclusions and Ethical Implications:

**Conclusions**

Our final analysis suggests that our two selected hallmark programs are biologically relevant in COAD, but they do not strongly track tumor stage on their own. In the unsupervised

analysis, proliferative-signaling scores and apoptosis-resistance scores showed some positive association across samples, but our stage-based boxplots overlapped substantially. PCA and UMAP also did not show strong separation of Stage I–IV tumors, and the best K-means solution had a low silhouette score of **0.154**. This indicated to us that there was a weak cluster structure in our 40-gene feature space.

In the supervised analysis, the improved logistic-regression model performed better than the baseline on the validation set and was therefore selected as the final model. When we applied this selected model to the held-out test set, its performance remained modest (**test AUROC = 0.589**, **test balanced accuracy = 0.547**). Because the test set was not used during model fitting or model selection, these values provide our final estimate of out-of-sample performance. Taken together, the validation and test results suggest that the two hallmark programs contain some stage-related information, but that signal is weak and not sufficient by itself to strongly predict simplified tumor stage in this dataset.

From a modeling perspective, our results highlight a key lesson: biological relevance does not guarantee predictive power. While the selected hallmark genes are mechanistically meaningful, they capture only a subset of the complexity driving tumor progression. Tumor stage reflects not only intrinsic tumor biology but also interactions with the tumor microenvironment, immune system, and broader systemic factors that are not captured in a 40-gene panel. Our findings are also consistent with the broader cancer-biology literature in suggesting that hallmark-associated expression programs are biologically informative, but they are not by themselves sufficient to fully explain a complex clinical phenotype such as tumor stage.

**Ethical Implications**

In an ethical sense, these findings carry important ethical considerations for the use of machine learning in biomedical contexts. First, they emphasize that models built on limited datasets and showing only modest held-out test performance should not be used for clinical decision-making without extensive external validation. Misclassification of tumor stage (particularly underestimating late-stage disease) could lead to inappropriate treatment decisions and harm to patients.

Second, our dataset represents a subsampled and potentially non-representative cohort, meaning that learned patterns may not generalize across diverse patient populations. If such models were deployed without addressing these biases, they could contribute to health disparities, especially for underrepresented groups in genomic datasets.

Finally, our work highlights the importance of interpretability and transparency. While more complex models could potentially improve predictive performance, simpler models like logistic regression allow us to understand which biological features are contributing to predictions. In a clinical setting, this interpretability is critical for building trust and ensuring that models are used responsibly.

Overall, our study supports the idea that hallmark-based gene expression analysis is valuable for understanding tumor biology, but insufficient on its own for clinical prediction.

# Limitations and Future Work:

**Limitations**

Several limitations in our study likely contributed to the modest predictive performance and should be considered when interpreting our results:

1. Our analysis focused on a 40-gene hallmark panel, which, while biologically interpretable, captures only a small fraction of the molecular complexity of COAD. Important pathways such as immune response, angiogenesis, metabolism, and genome instability were not included.
2. The training dataset contains a relatively small number of COAD samples which limits statistical power and increases the risk of overfitting. Additionally, variability in metadata completeness and tumor annotation introduces noise into the analysis.
3. We reduced tumor stage into a binary classification (Early vs. Late), which may oversimplify meaningful biological differences between stages. This likely reduces the model's ability to detect more nuanced relationships.
4. Our evaluation does not fully test generalizability to independent datasets or clinical populations.
5. Our model relies solely on RNA-expression data, ignoring other important data types such as DNA mutations, copy-number variation, epigenetics, and proteomics, all of which contribute to tumor behavior.

**Future Work**

To build on our findings, several improvements could strengthen both the biological insight and predictive performance of this analysis:

1. Incorporating additional cancer hallmarks (e.g., angiogenesis, immune evasion, genome instability) could provide a more complete representation of tumor biology and improve model performance.
2. Instead of relying on individual gene expression, applying methods such as Gene Set Variation Analysis or enrichment-based scoring could better capture coordinated pathway activity.
3. Combining RNA expression with mutation data (e.g., KRAS, TP53 status), copy-number alterations, or methylation profiles would allow for a more comprehensive model of tumor progression.
4. Predicting full tumor stage (I–IV) or clinically relevant outcomes such as survival or recurrence may provide a more meaningful assessment of biological relevance.
5. Expanding to larger datasets or combining multiple cohorts would improve generalizability and reduce bias, especially across different demographic and clinical

populations.
6. Testing the model on independent datasets is critical before any real-world application.

# Notes:

- **Completed:**
  - Data and Disease Background: Dani and Will
  - Data Analysis: Dani and Will
  - Verification/Validation: Dani and Will
  - Conclusions/Ethics: Will
  - Limitations/Future Work: Dani
- **To Do:**
  - Nothing at this time.
- **Questions/Concerns:**
  - Nothing at this time.
- **References:**

  - [1] S. Groves and S. Peirce-Cottler, "Cancer Hallmarks ABM – Quick Reference," course handout, Module 4, Computational Biomedical BME 2315, University of Virginia, 2026.
  - [2] M. Rahman et al., "Alternative preprocessing of RNA-Sequencing data in The Cancer Genome Atlas leads to improved analysis results," Bioinformatics, vol. 31, no. 22, pp. 3666–3672, 2015, doi: 10.1093/bioinformatics/btv377.
  - [3] National Center for Biotechnology Information, "GSE62944 - GEO Accession viewer," Gene Expression Omnibus. [Online]. Available: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE62944.
  - [4] National Cancer Institute, "Colon Cancer Treatment (PDQ®)–Patient Version." [Online]. Available: https://www.cancer.gov/types/colorectal/patient/colon-treatment-pdq
  - [5] American Cancer Society, Cancer Facts & Figures 2026. Atlanta, GA, USA: American Cancer Society, 2026.

- **AI Statement:**
  - External use of LLMs was utilized on a case-by-case basis, often for initial source discovery + inspiration and code debugging. These resources were very great for finding research done on particular Hallmarks and gene sets. GitHub Copilot was especially helpful with providing 'Inline Suggestions' for shortening time with boilerplate code and repetitive function naming. This was particularly helpful for

implementing LaTeX format into our markdown segments to that we can easily see the equations being used in a clear sense. However, the final code structure, modeling choices, and interpretations were based on the course lectures, GitHub materials, and provided resources, with all final analysis completed by our team.

## QUESTIONS FOR YOUR TA:

We have no questions at this time.